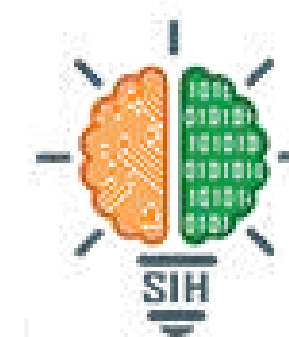


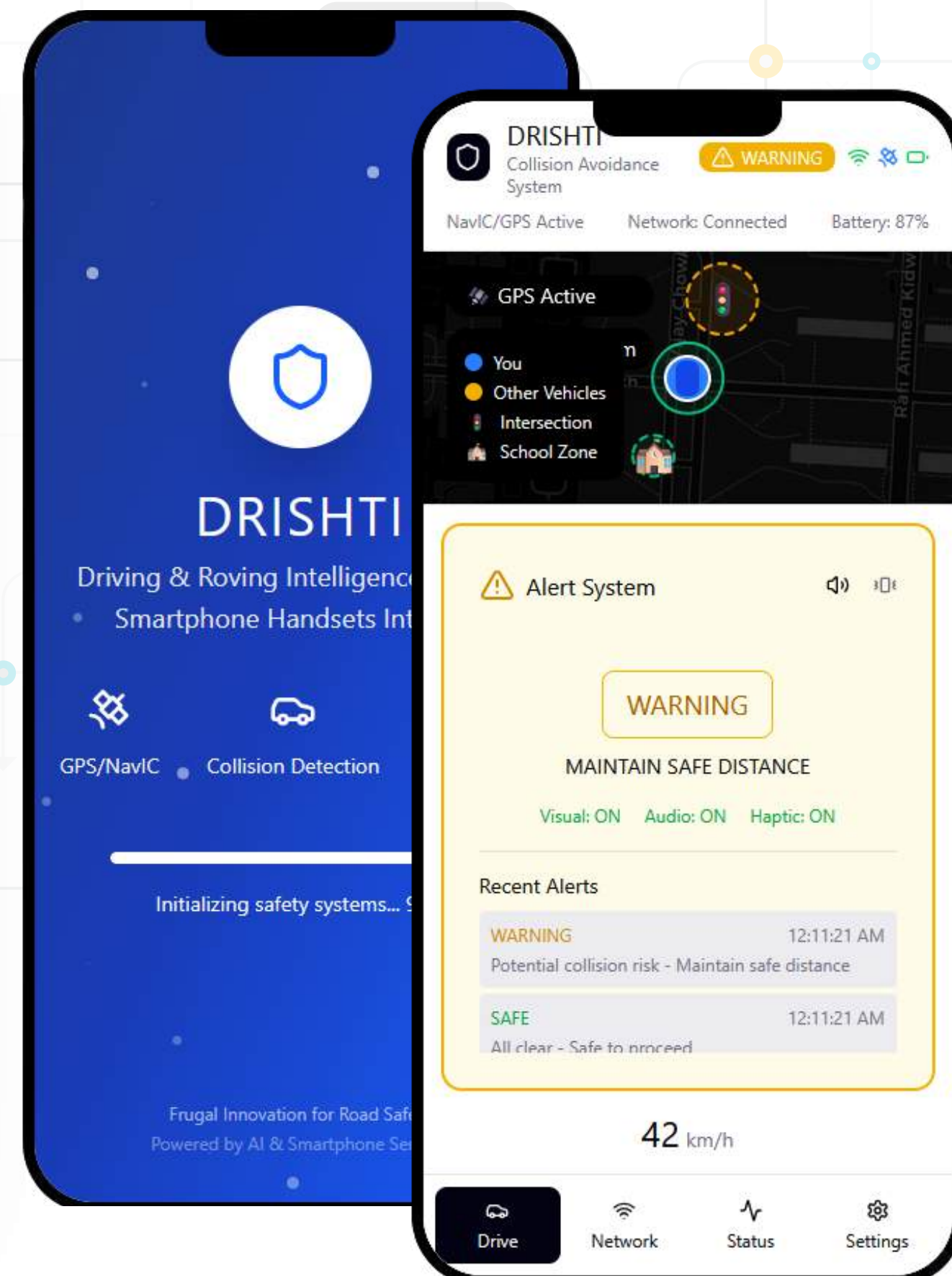
SMART INDIA HACKATHON 2025



SMART INDIA
HACKATHON
2025

TITLE PAGE

- Problem Statement ID – SIH25177
- Problem Statement Title– Use of measurements from the mobile phones (low cost preferred) to provide a safe autonomous navigation on the roads
- Theme– Smart Vehicles
- PS Category– Software
- Team ID– 62019
- Team Name : Data Morphers





DRISHTI (Driving & Roving Intelligence through Smartphone Handsets Interface)

An Android application that transforms a network of smartphones into a collaborative, decentralized anti-collision system.

DRISHTI directly replaces expensive LiDAR/radar with intelligent software.

It creates a real-time "safety bubble" between vehicles using only the sensors already in their phones.

It fuses raw data from three sources: ISRO's NavIC, the phone's IMU (motion sensors), and the Camera.

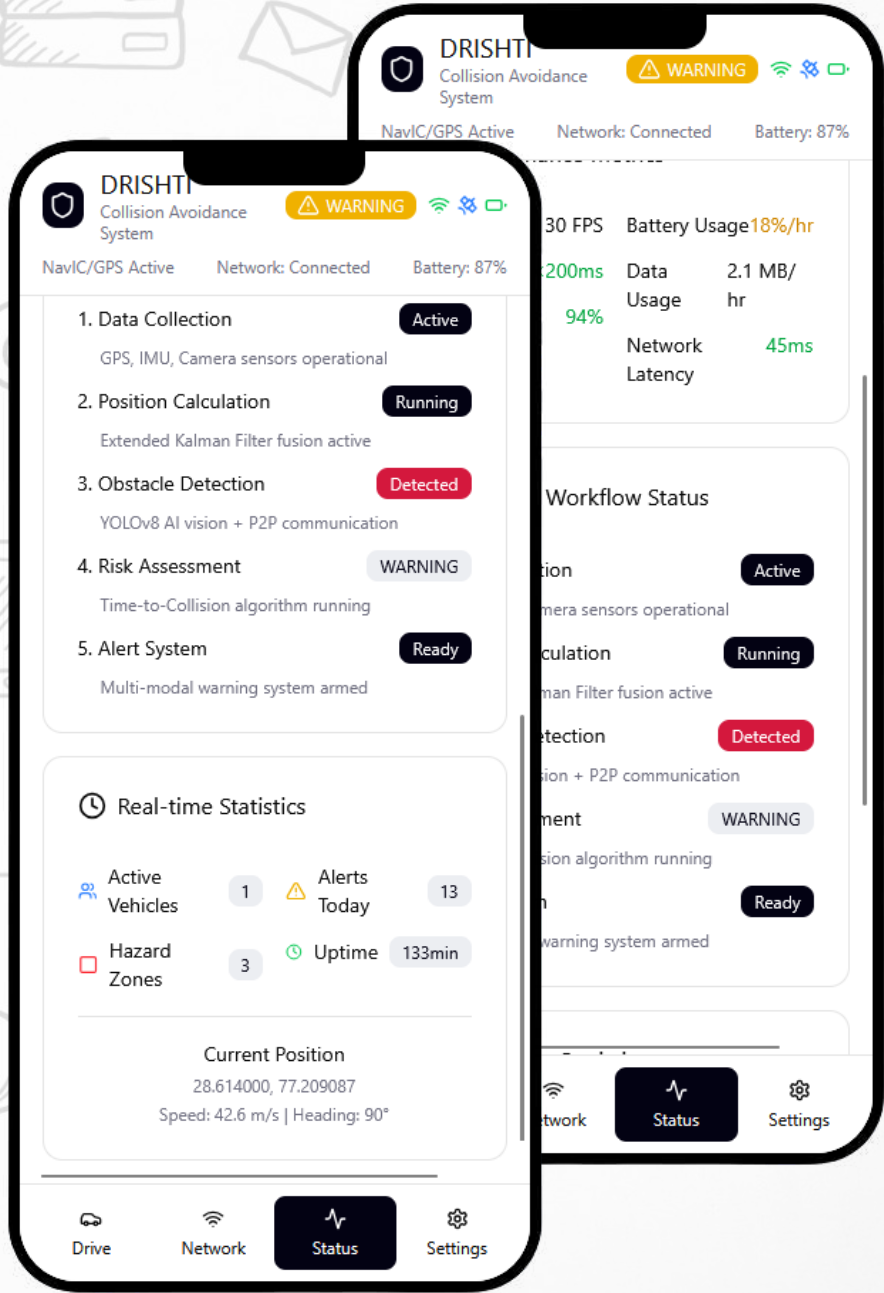
Cooperative Positioning: The system's accuracy improves as more users join the network, compensating for errors on low-end phones.

Vehicles communicate directly via a Peer-to-Peer (P2P) Wi-Fi Direct network, eliminating the need for a central server.

Advanced Sensor Fusion: A unique combination of NavIC + IMU + AI-powered Computer Vision provides robust tracking where GPS alone would fail.

India-Centric: Prioritizes ISRO's NavIC constellation for superior positioning accuracy and reliability in the Indian region.

An on-device AI model (YOLOv8) uses the camera to detect non-connected obstacles (pedestrians, other cars).



TECHNICAL APPROACH



MOBILE PLATFORM

- Android:** Recommended for full sensor access (Java+Kotlin)
-  **Mapping & Navigation**
- Google Maps SDK:** Maps, markers, routes, hazard points
- Optional: Mapbox SDK:** Custom maps, lightweight alternative

SENSOR & DEVICE APIS

- GPS / Location APIs → Latitude, longitude, speed, heading
- Accelerometer & Gyroscope → Detect sudden braking, lane changes
- Magnetometer → Compass/heading accuracy
- Camera → Optional for obstacle detection (YOLOv8n / MobileNet SSD Lite)

AI / OBSTACLE DETECTION (OPTIONAL)

- TensorFlow Lite → On-device lightweight ML models
- PyTorch Mobile → Alternative mobile AI deployment
- YOLOv8n Tiny → Real-time vehicle/object detection
- MobileNet SSD Lite → Lightweight detection for low-power devices

REAL-TIME DATA SHARING & COOPERATIVE POSITIONING

- Firestore Realtime Database:** Real-time vehicle positions
- Firestore Firestore:** Cloud storage of trip data & logs
- WebSockets / Socket.IO:** Low-latency alerts & data exchange



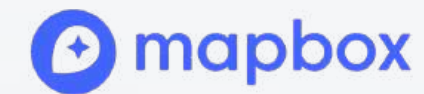
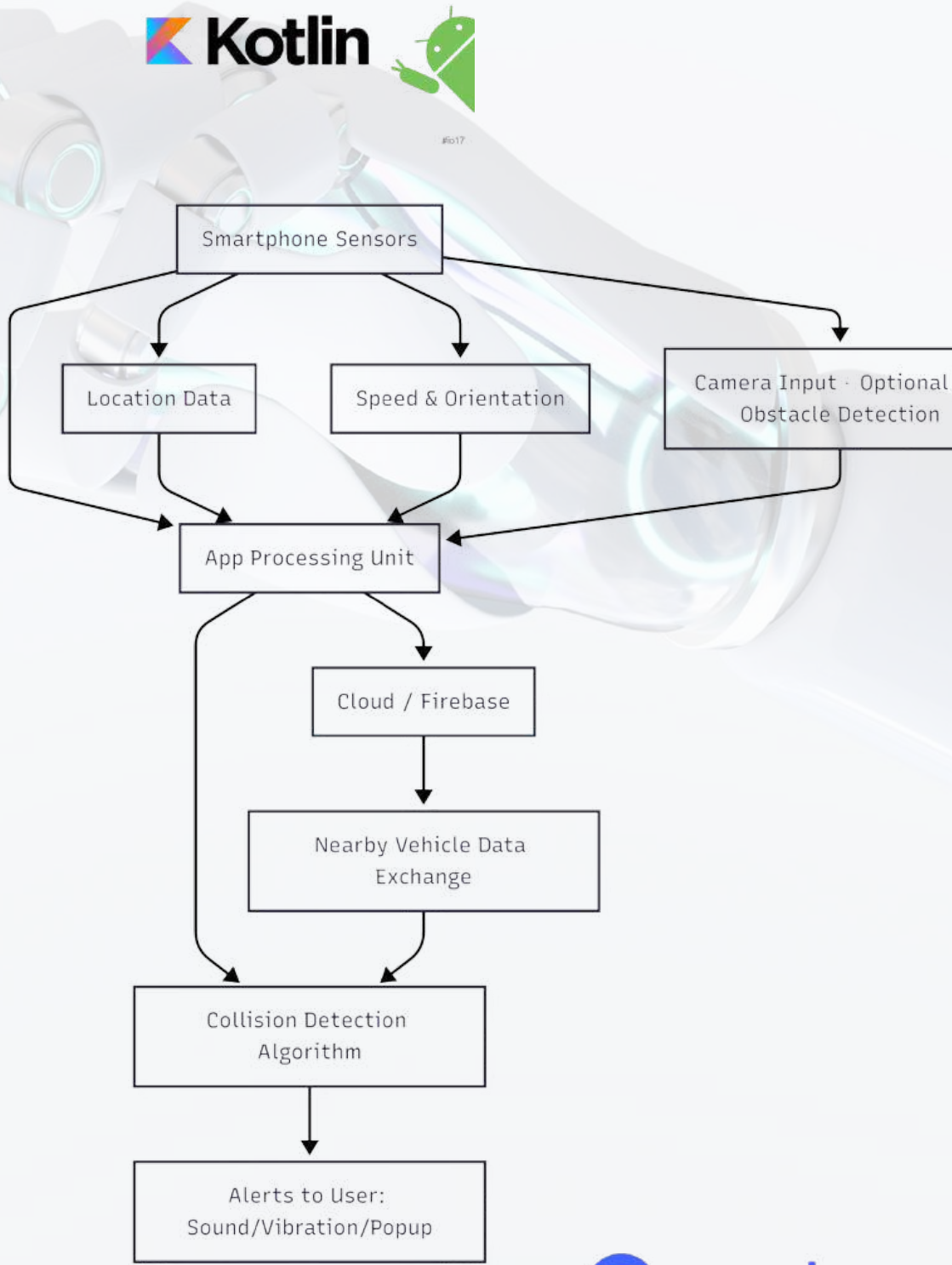
ALERTS & NOTIFICATION

- Android Notifications API:** Sound, vibration, popup alerts
- In-app UI elements:** Map markers, blinking alerts, dashboard warnings



APP ARCHITECTURE / FRAMEWORK

- MVVM (Model-View-ViewModel) → Clean Android app architecture
- Repository Pattern → Data management between sensors & cloud
- Dependency Injection (Hilt / Dagger) → Better modularity



FEASIBILITY AND VIABILITY

Technical Feasibility

Smartphone Sensors: GPS/GLONASS/NavIC for positioning; IMU (accelerometer, gyroscope) for movement & orientation; cameras for lane and obstacle detection.

AI & Software: Lightweight object detection models (YOLOv8n, MobileNet SSD) run in real-time on mobile.

Navigation APIs: Google Maps / Mapbox for blind-turns, intersections, and traffic alerts.

- **Deployment:** Simple app + dashboard mount; pilot tests possible with a few vehicles.
- **Indoor Mode:** Feasible via Bluetooth/Wi-Fi/UWB for parking lots or garages.

Smartphone Sensor
GPS,IMU,Camera..

Data Fusion and AI Models
YOLOv8n,Kalman Fi.

Navigation and Positioning
Google Maps/ Mapbox

Communication and Sharing
Wi-Fi Direct/ Bluetooth

Collision Detection and
Alerts

Technical Feasibility

- 1-2m accuracy
- Real-time AI
- Indoor support

Economic Viability

- Cost Rs 300-500
- Works on all vehicles
- 750M+ users base

Economic & Adoption Viability

Cost Advantage: Smartphone + mount (~₹300-500) vs ADAS hardware (₹1-3 lakh).

Ease of Adoption: Works for cars, two-wheelers, EVs, taxis, fleets; no retrofitting needed.

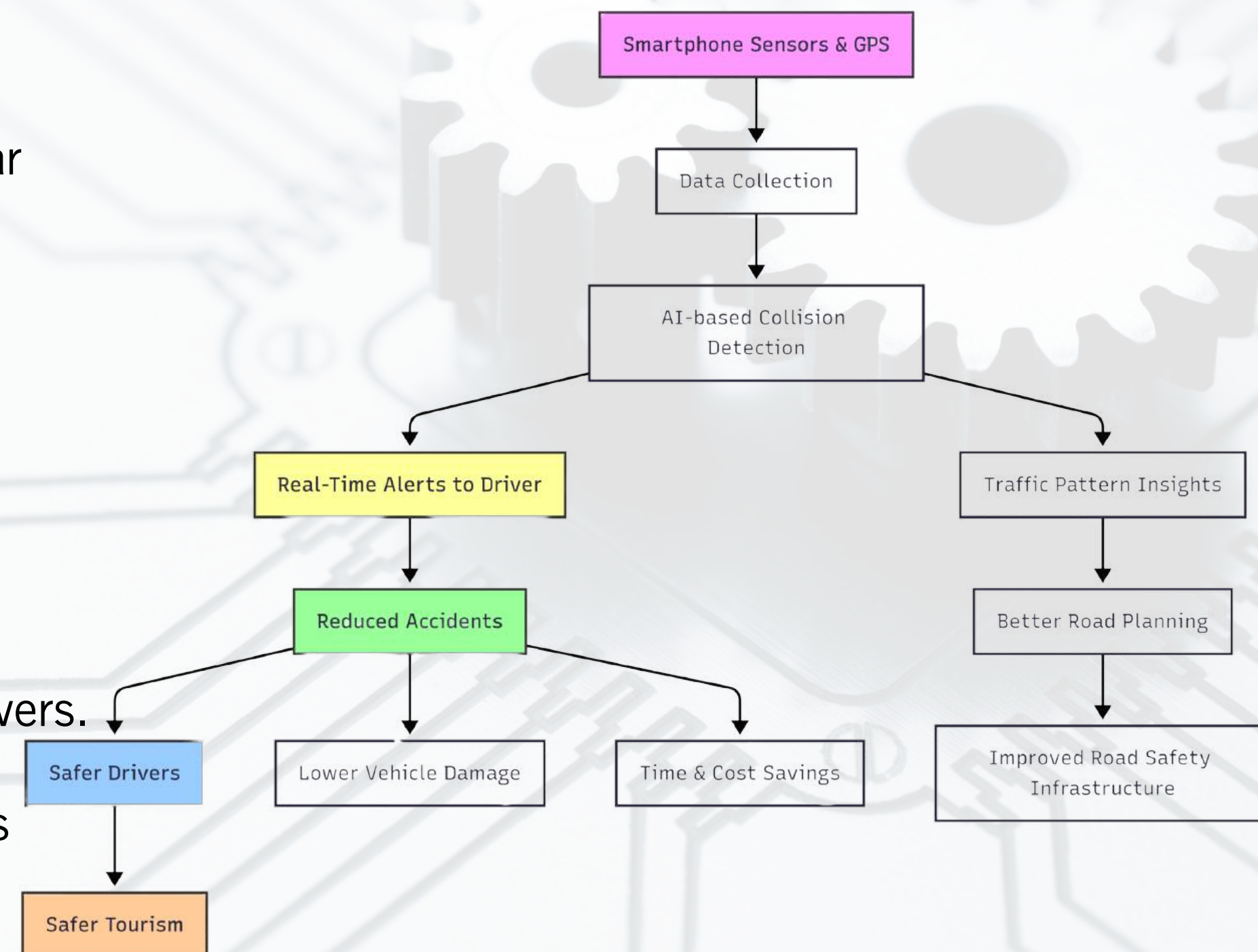
- **Business Model:** Freemium app with premium features (AI driver scoring, insurance discounts).
- **Scalability:** Cloud + P2P architecture; supports millions of users; future-ready with 5G/V2X integration.

Impact

- Reduce accidents in highways, cities, rural roads
- Affordable safety for common people, not just luxury car owners
- Step toward smart, connected mobility in India
- Supports Digital India & Atmanirbhar Bharat goals

Benefits

- Low cost → just a smartphone & mount
- Works on all cars, bikes, EVs
- Easy adoption → no hardware upgrade
- Cooperative Network → Safer roads with connected drivers.
- Enhances driver awareness without distraction
- Potential for integration with govt. road safety programs



RESEARCH AND REFERENCES

Details / Links of the reference and research work:

Official Government & Policy Reports

Ministry of Road Transport & Highways (MoRTH), Govt. of India
Source: Road Accidents in India - 2023 Report

NITI Aayog
Reports on Smart Mobility and Urban Transport Initiatives.

Core Technical Documentation

Android Developer

Documentation (Google).
Accessing Raw GNSS
Measurements

Wi-Fi Direct (Wi-Fi P2P) API

ISRO Navigation System (NavIC)

Official NavIC / IRNSS Technical
Specifications & Performance
Reports.

Key Algorithms & Academic Concepts

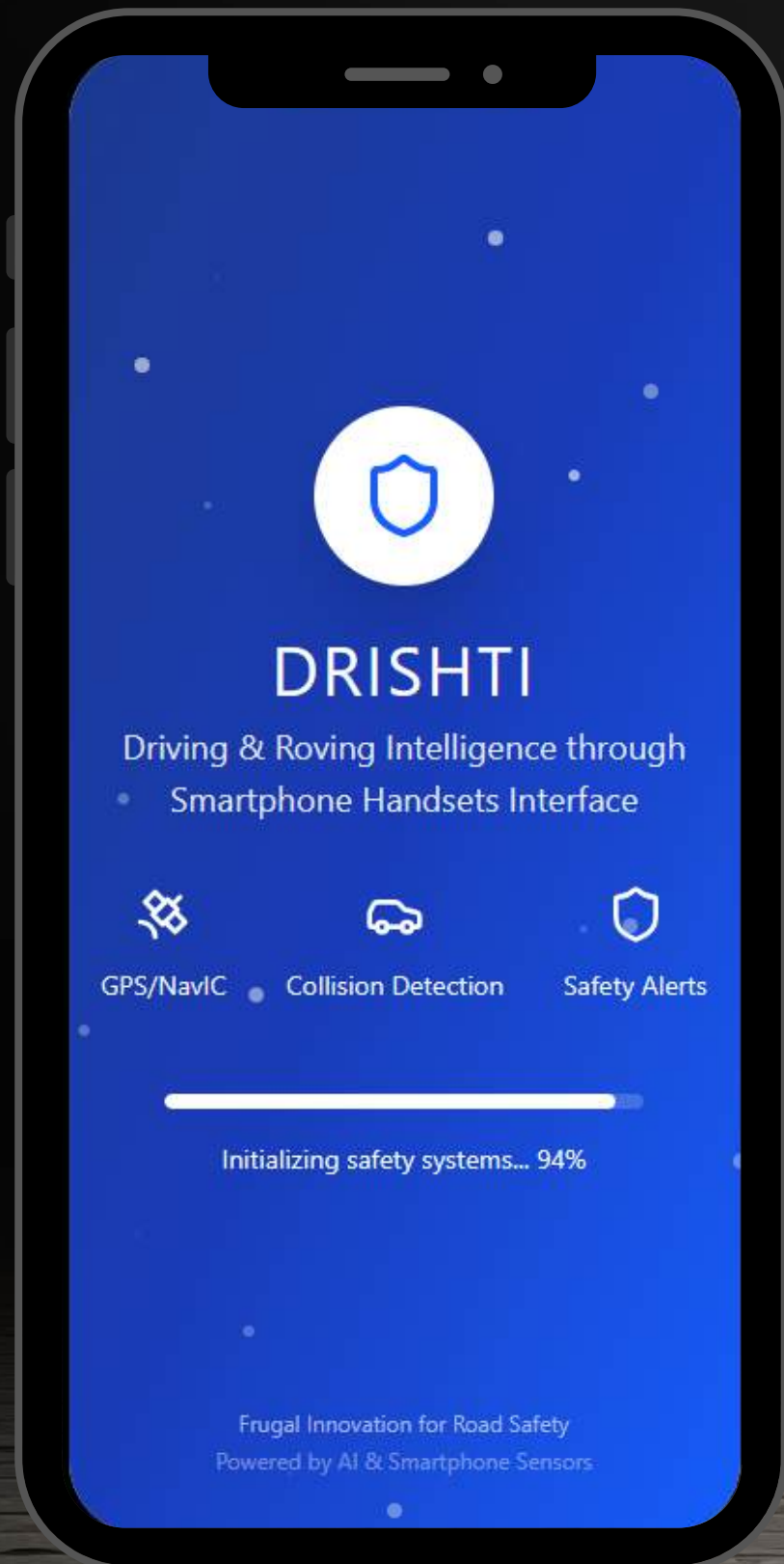
Sensor Fusion (Kalman Filter)

Source: "An Introduction to the
Kalman Filter" by Greg Welch &
Gary Bishop, University of North
Carolina.

AI on Mobile (Object Detection)

TensorFlow Lite

YOLOv8



DRISHTI

(Driving & Roving Intelligence through Smartphone Handsets Interface)

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SMART INDIA
HACKATHON
2025

PROBLEMS



Democratizing advanced safety – turning every smartphone into a collision avoidance system

1

Advanced safety systems remain a luxury, not a standard.

2

Two-wheelers: 75% of Indian traffic, 0% collision protection

3

Vehicles operate in data silos no communication, no awareness.

4

The Latency Challenge in Real-Time Decision Making

5

Navigation systems fail where we need them most: indoors.

6

1Hz GPS updates can't match 30Hz collision dynamics

Road Accidents in India – 2023 (MoRTH Official Report) : Source

National Overview

Metric	Value (2023)
Total Road Accidents	480,583
Total Persons Killed (Fatalities)	172,890
Total Persons Injured	462,825
Total Fatal Accidents	160,509
Accident Severity (Deaths per 100 accidents)	36

Daily & Hourly Averages

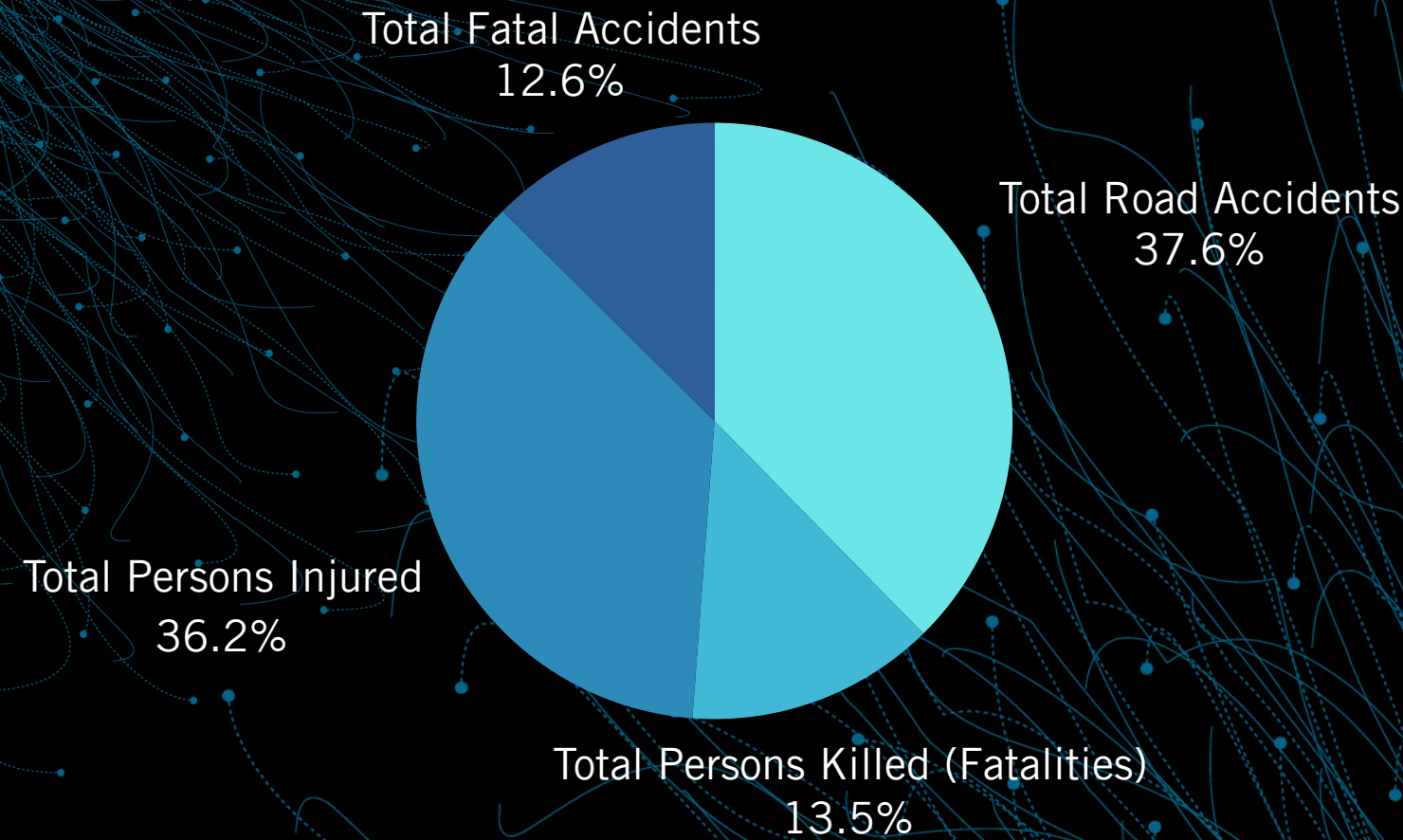
Metric	Value
Accidents per day	1,317
Deaths per day	474
Accidents per hour	55
Deaths per hour	20

Comparison with 2022

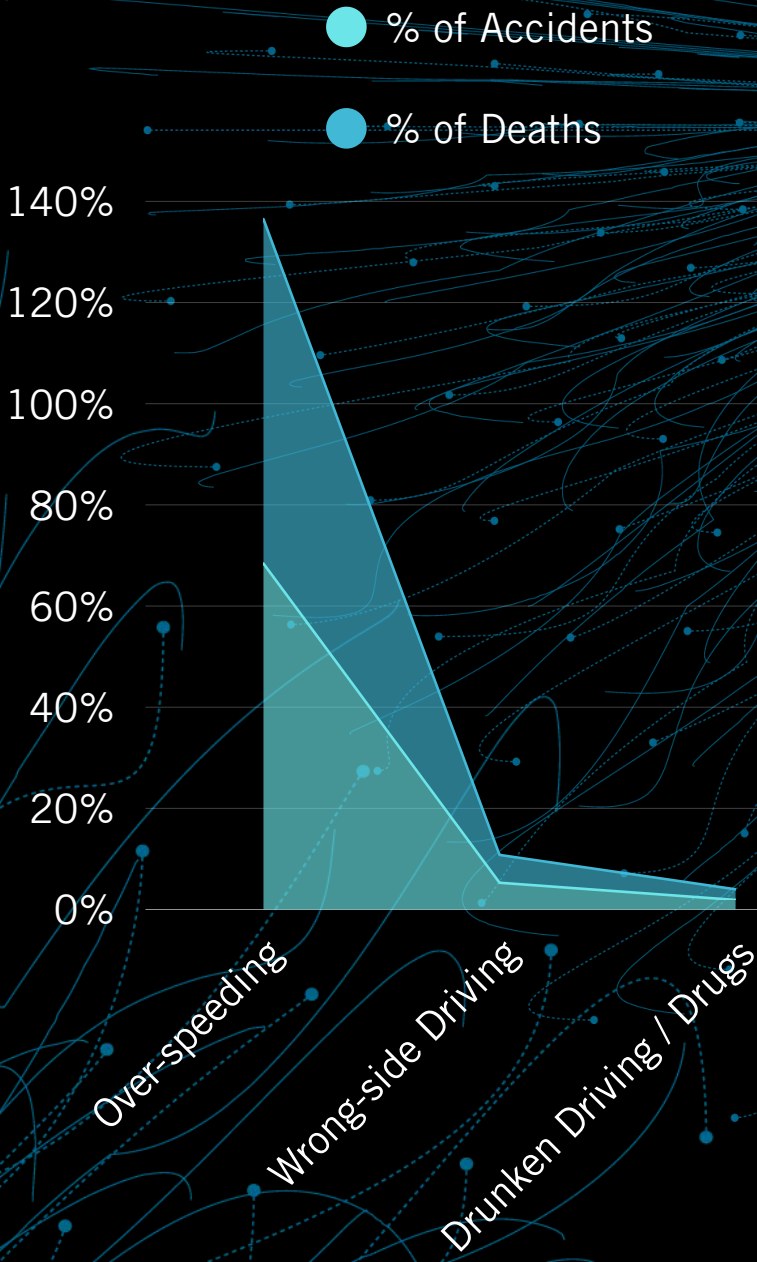
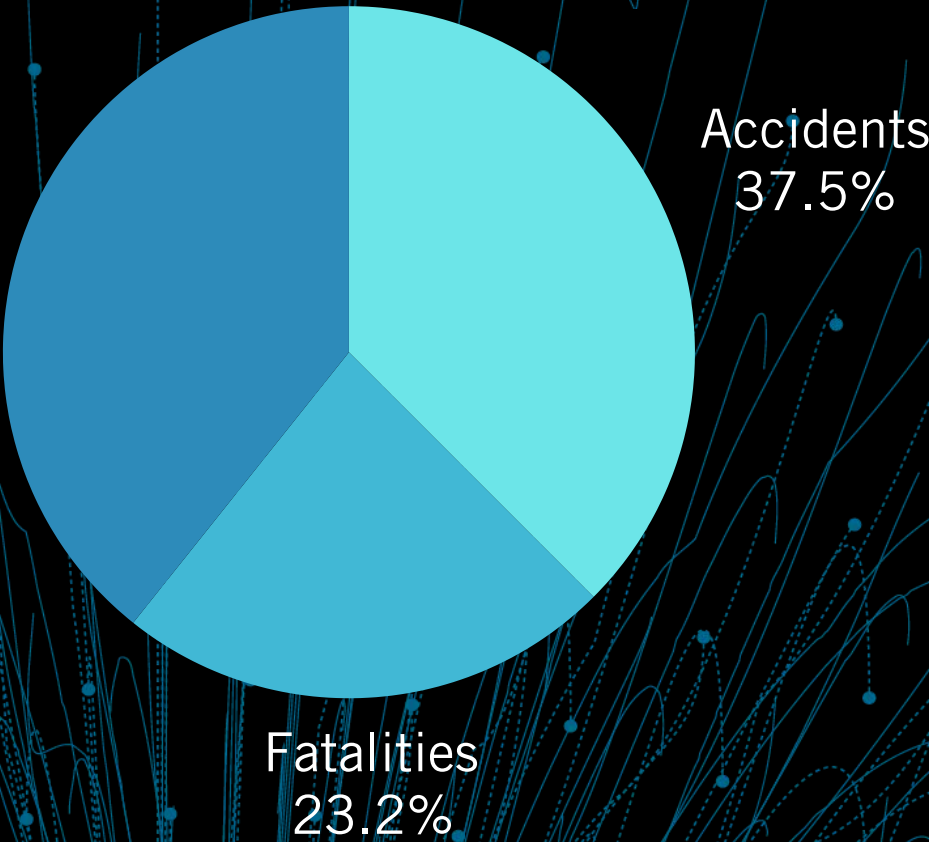
Category	% Change
Accidents	4.20%
Fatalities	2.60%
Injuries	4.40%

Primary Causes of Accidents

Cause	% of Accidents	% of Deaths
Over-speeding	68.40%	68.10%
Wrong-side Driving	5.30%	5.50%
Drunken Driving / Drugs	1.90%	2.10%



Injuries
39.3%



Solution - A Software-Only Revolution in Vehicle Safety

1 100% smartphone-based - zero additional hardware required

2 Sensor fusion magic: GPS + IMU + Camera = Super-accurate positioning

3 GPS and GNSS usage for latitudnal and longitudnal positioning .

4 Your phone's camera becomes a virtual LiDAR with AI vision

5 Vehicles talk directly to each other via Wi-Fi Direct.

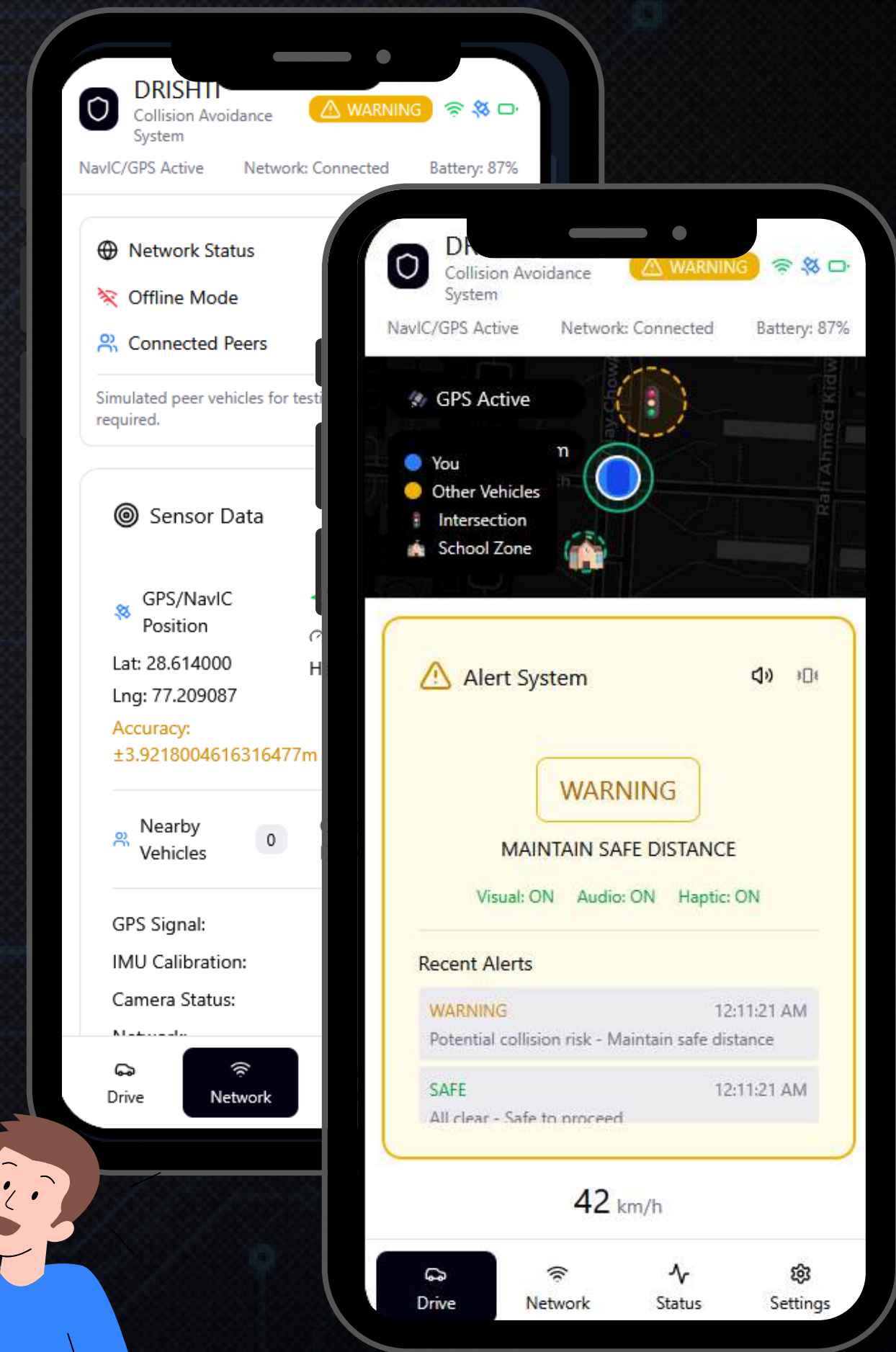
6 See around corners through peer vehicle data sharing

7 Predict collisions before they happen with Time-to-Collision algorithms

8 Seamless indoor-outdoor navigation using phone anchors.

9 Smart fallbacks keep you protected even when signals drop

10 The more users, the smarter the system gets for everyone.



IMPACTS

Road Safety Innovation

Phones as low-cost safety devices for EVs, driverless cars, and pedestrians.

Reduces collisions at intersections

Addresses blind spots

Improves safety on busy roads

Affordable & Scalable

No need for costly LiDAR, radar, or ADAS hardware.

Works on everyday smartphones

Scalable to millions quickly

Low implementation cost

Smart Mobility Infrastructure

Enables V2V and V2I communication without extra hardware.

Vehicle-to-vehicle (V2V)

Vehicle-to-infrastructure (V2I)

Foundation for C-ITS

Indoor + Outdoor Coverage

GPS + indoor anchor tracking for comprehensive safety coverage.

Roads & tunnels

Malls & warehouses

Parking lots

Data-Driven Insights

Logged data helps governments & companies improve safety.

Identify accident-prone zones

Improve traffic flow

Position errors & latency analysis

BENEFITS

For Users (Drivers, Riders, Pedestrians)
Real-time alerts (visual, audio, vibration) → safer driving & confidence.
Works in poor visibility and congested traffic.

For EV & Autonomous Vehicle Ecosystem
Provides low-cost collision warning.
Improves safety perception, encouraging wider EV & driverless adoption.

Social & Environmental
Fewer accidents → less congestion → lower emissions.
Boosts India's global leadership in intelligent transport innovation.

For Governments & Smart Cities
Supports Vision Zero (road safety goals).
Lowers accident rates → reduces healthcare, insurance & emergency costs.

For Fleet Operators & Logistics
Prevents downtime & repairs.
Improves delivery reliability by cutting accident risks.



Real-Time Data Sharing & Cooperative Positioning

Firebase Realtime Database : Real-time vehicle positions
Firebase Firestore : Cloud storage of trip data & logs
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Maps & Navigation

Google Maps SDK: Maps, markers, routes, hazard points
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AI / Obstacle Detection

TensorFlow Lite : On-device lightweight ML models
PyTorch Mobile: Alternative mobile AI deployment
YOLOv8n Tiny: Real-time vehicle/object detection
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Alerts & Notifications

Android Notifications API: Sound, vibration, popup alerts
In-app UI elements: Map markers, blinking alerts, dashboard warnings

App Architecture / Frameworks

MVVM (Model-View-ViewModel) : Clean Android app architecture
Repository Pattern : Data management between sensors & cloud
Dependency Injection (Hilt / Dagger): Better modularity

Sensors & Device APIs

GPS / Location APIs : Latitude, longitude, speed, heading
Accelerometer & Gyroscope : Detect sudden braking, lane changes
Magnetometer : Compass/heading accuracy
Camera Optional for obstacle detection (YOLOv8n / MobileNet SSD Lite)

Mobile App (Java , Kotlin)

Modern Android development, Android SDK support

Technical Approach



Features



● System Health Monitoring

Shows GPS accuracy, network status, battery, and peer connectivity.



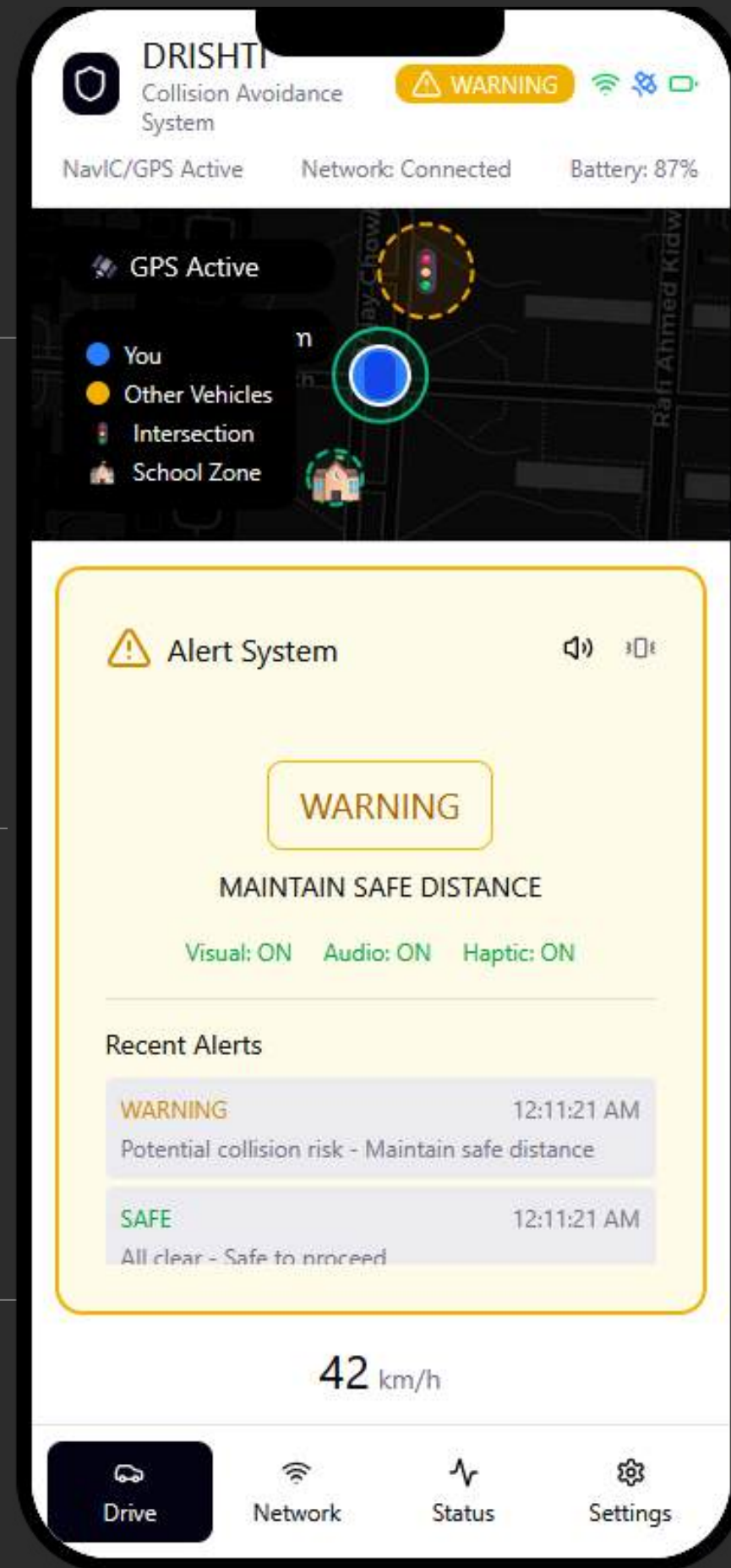
● Real-time Collision Detection

Continuously predicts potential vehicle collisions using TTC and predictive path analysis.



● Multi-Vehicle Tracking

Manages live state data of user, peers, and AI-detected obstacles.



● Dynamic Safety Distance



User-configurable safe distance (20–100m) for personalized risk thresholds.

● Multi-Modal Alerts



Combines visual, audio, and vibration alerts for maximum driver awareness.

● Customizable Alerts



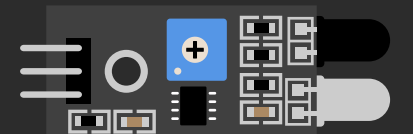
Users can toggle sound, vibration, and visual modes in settings.

● Progressive Warning System



Escalates from SAFE → WARNING → DANGER to avoid alert fatigue.

● Sensor Fusion Engine



Integrates GPS/NavIC, IMU, camera AI, and Wi-Fi/Bluetooth for accuracy.

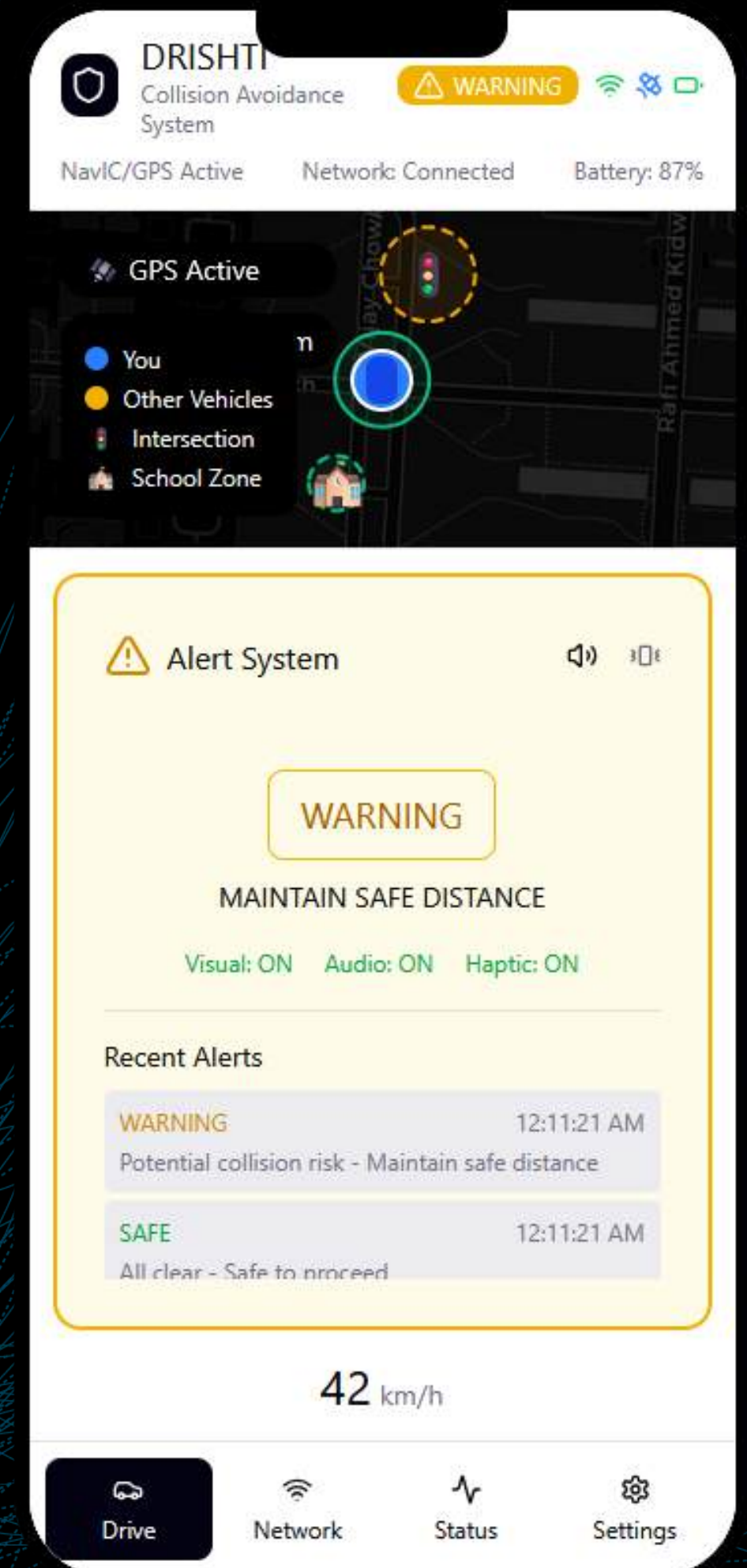
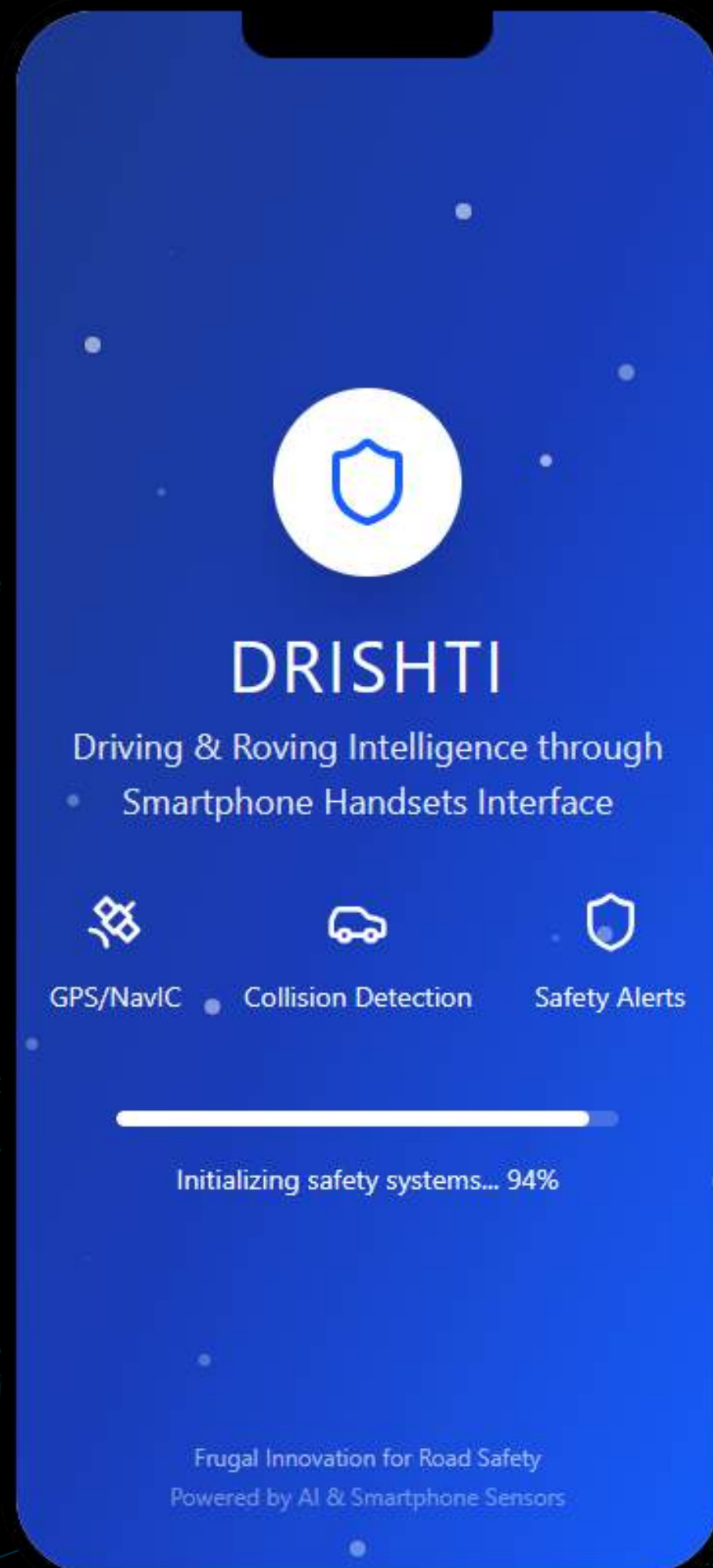
DRISHTI

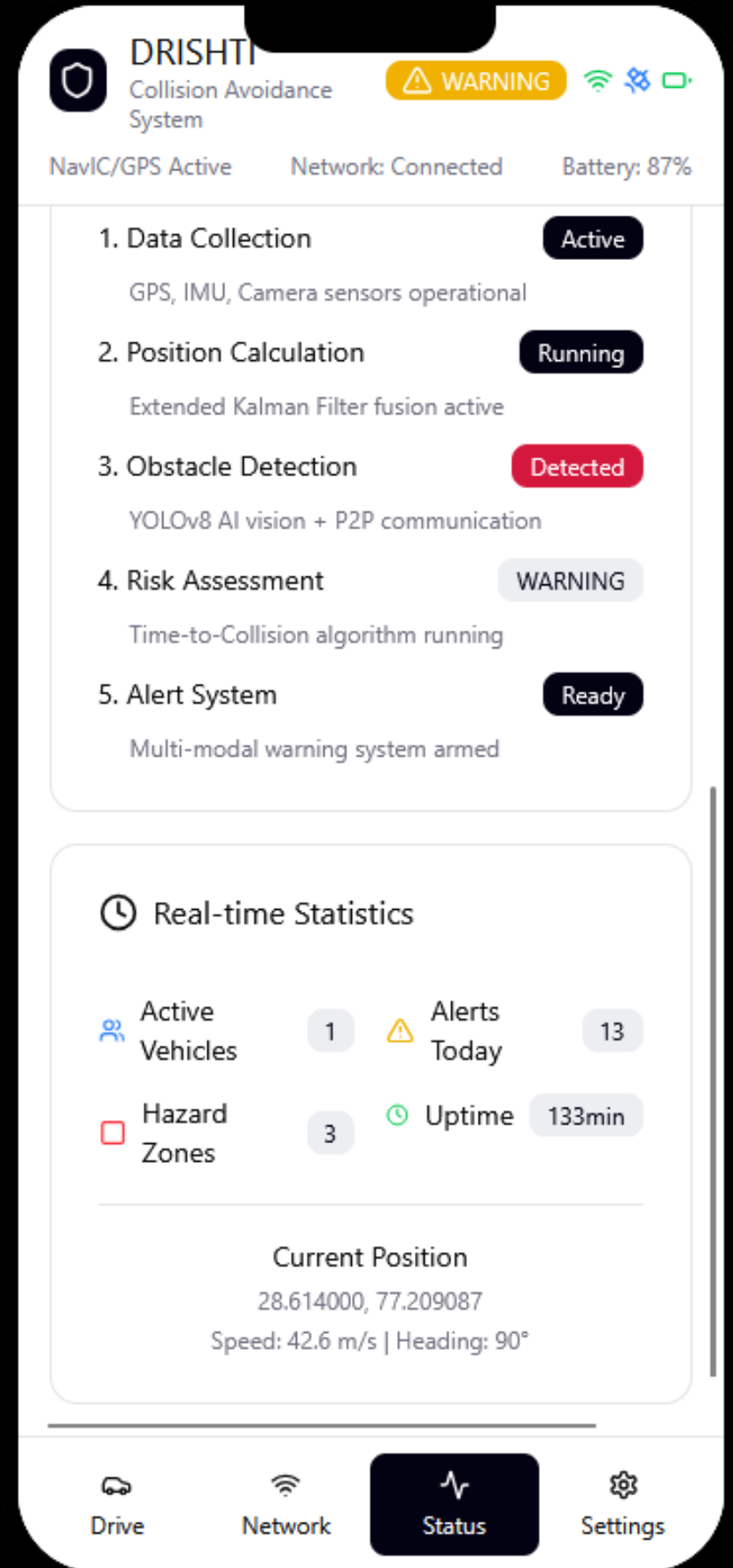
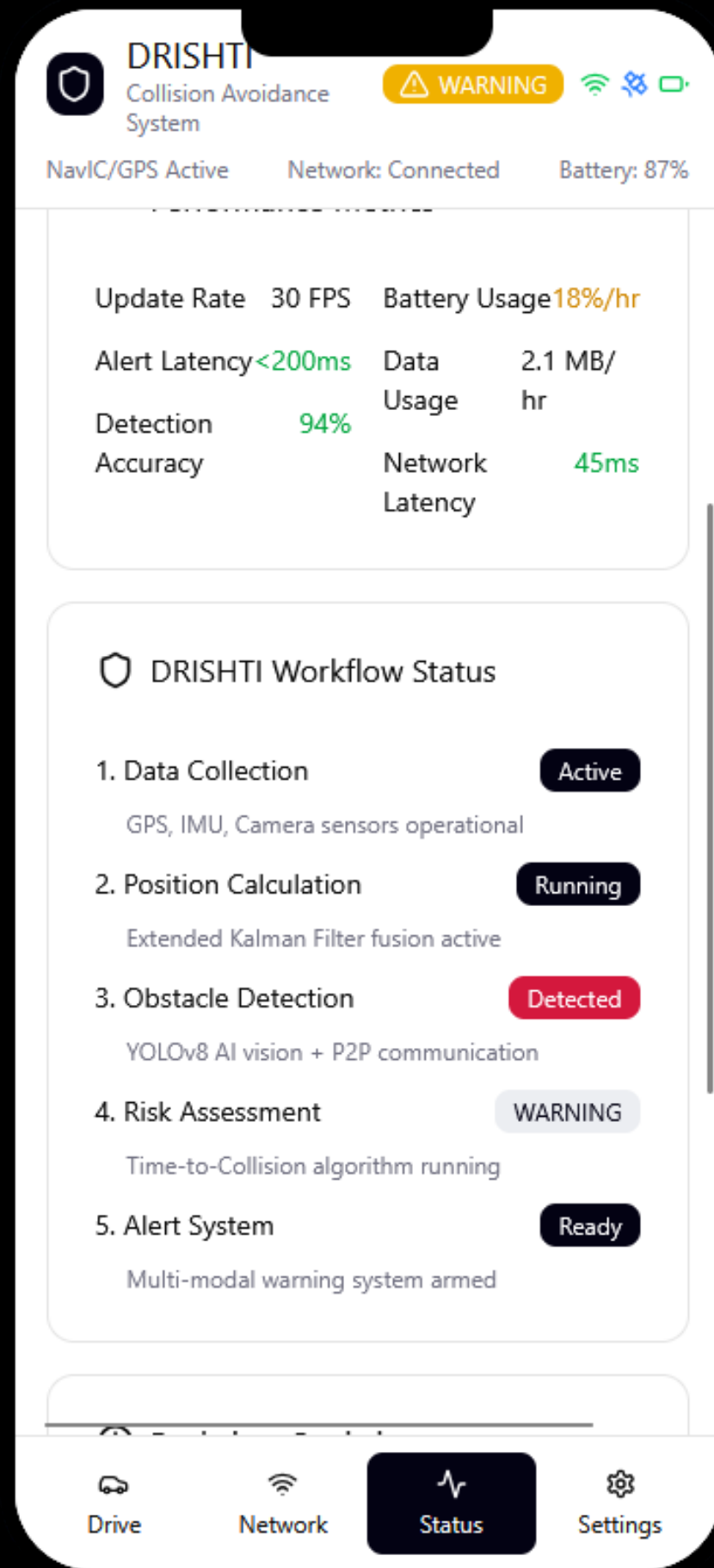
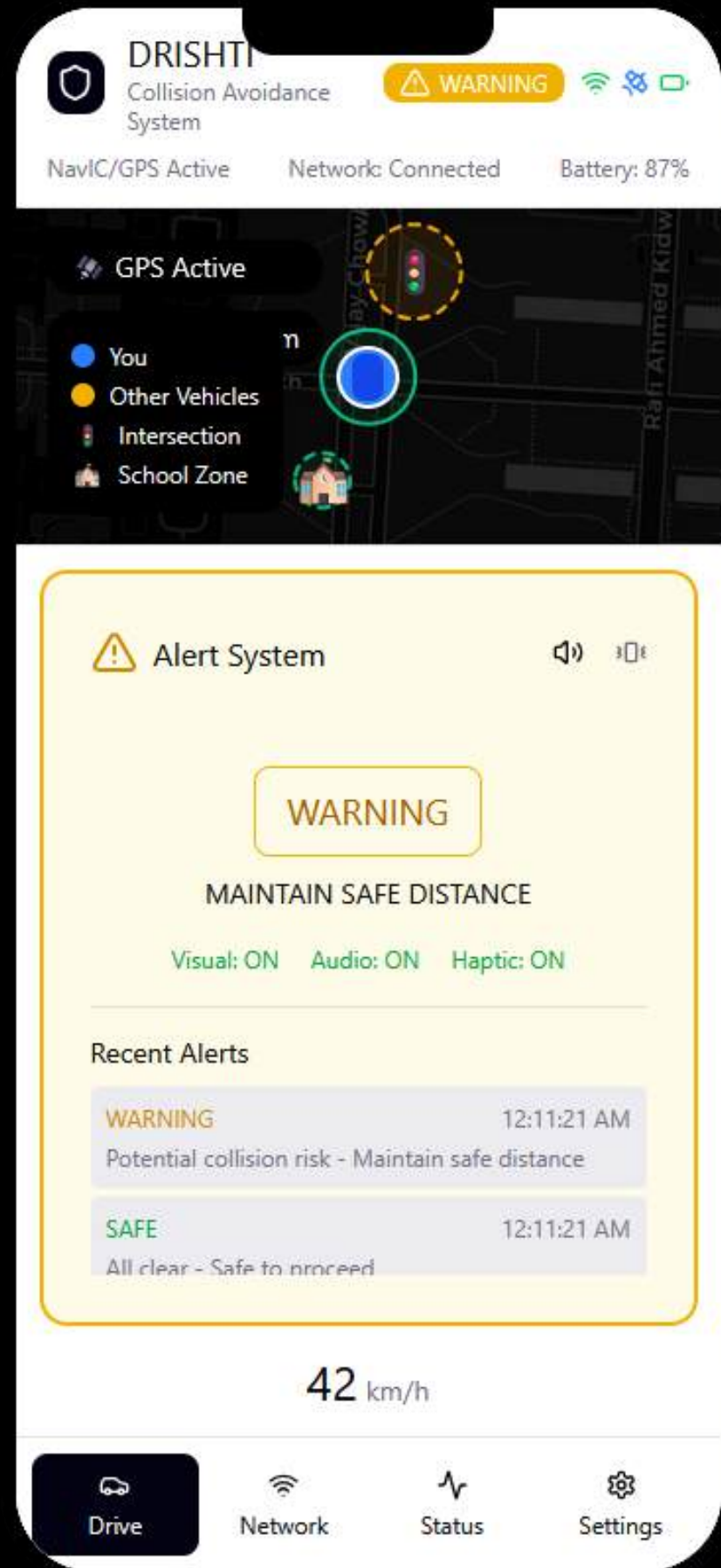
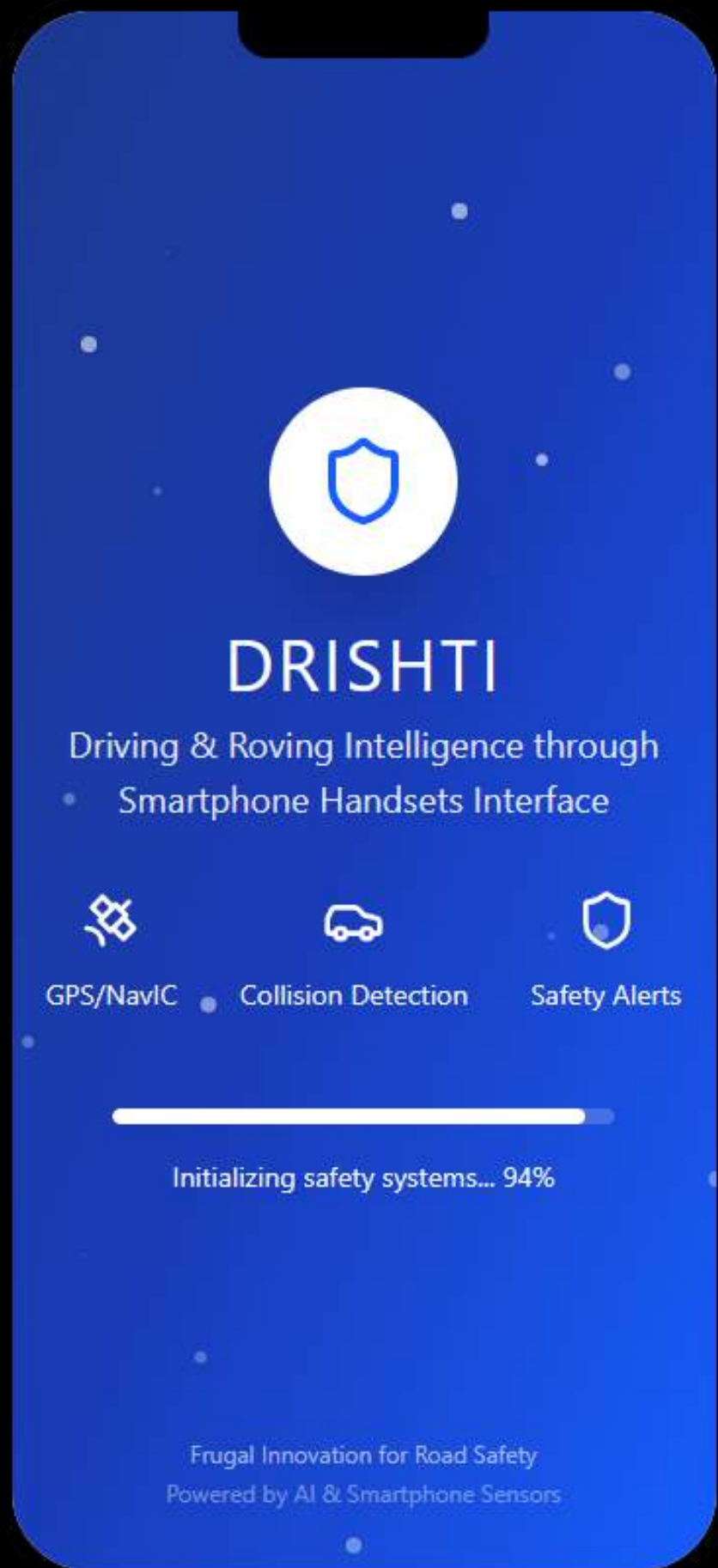
(Driving & Roving Intelligence through
Smartphone Handsets Interface)
Prototype

Integrated itinerary & real-time safety app for tourists.

Your smartphone can do what expensive
car systems can't for FREE

Turn your phone into a
collision avoidance system
zero hardware needed





Every phone turns into a chatty traffic buddy warning others, 'Hey, don't bump into me!'

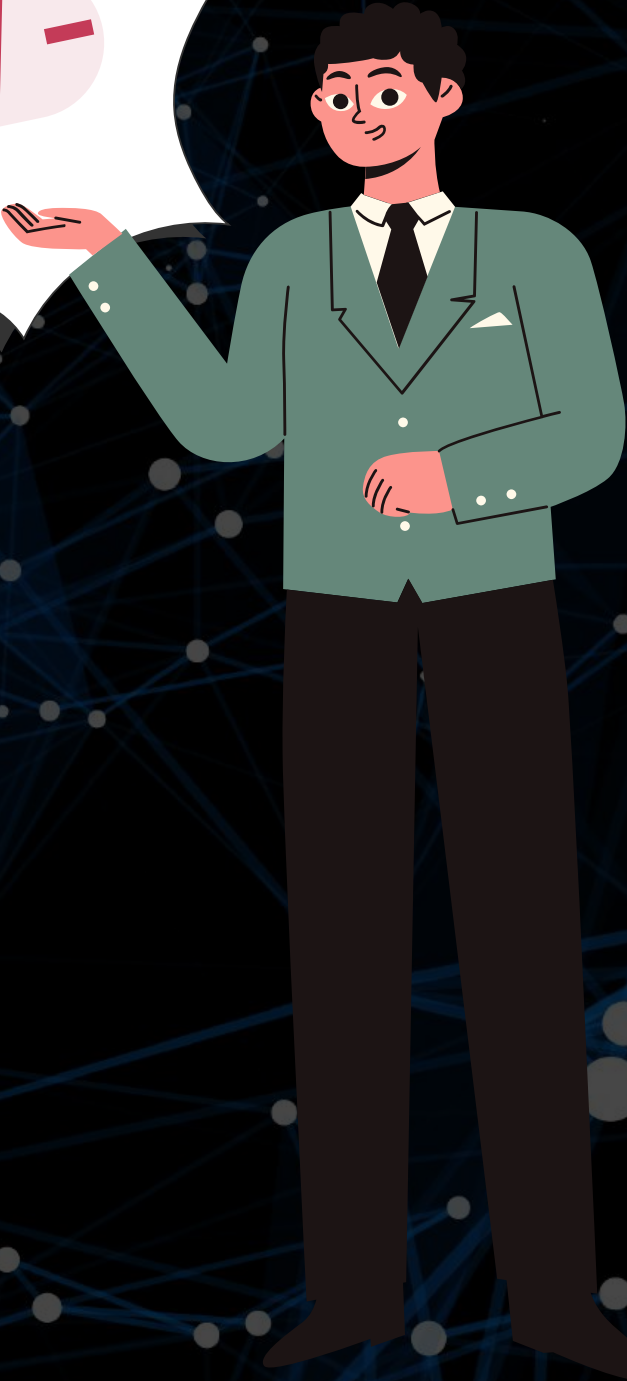


How much I have to pay for all this?



Unit Economics

Rs. 49/-



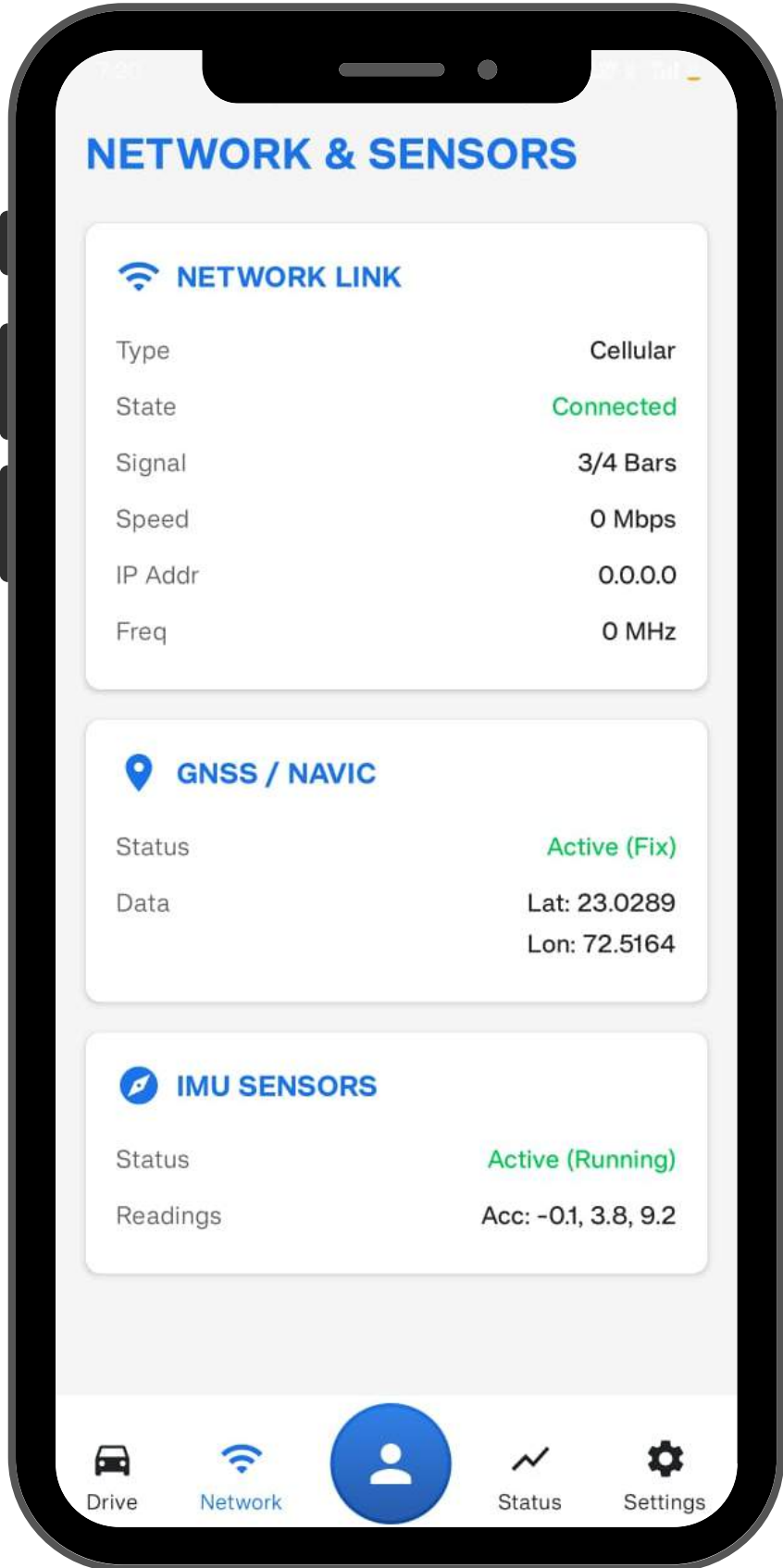
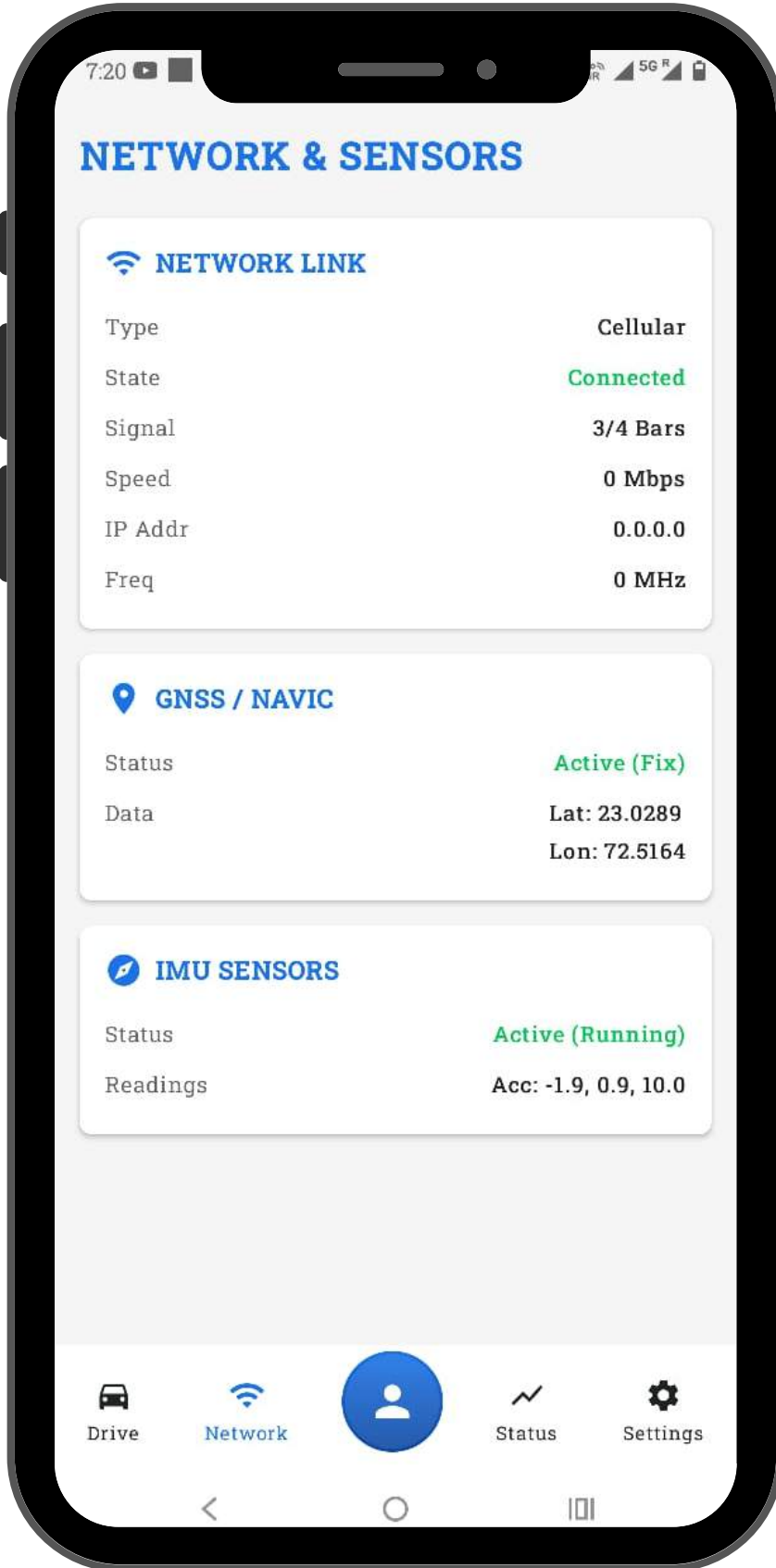
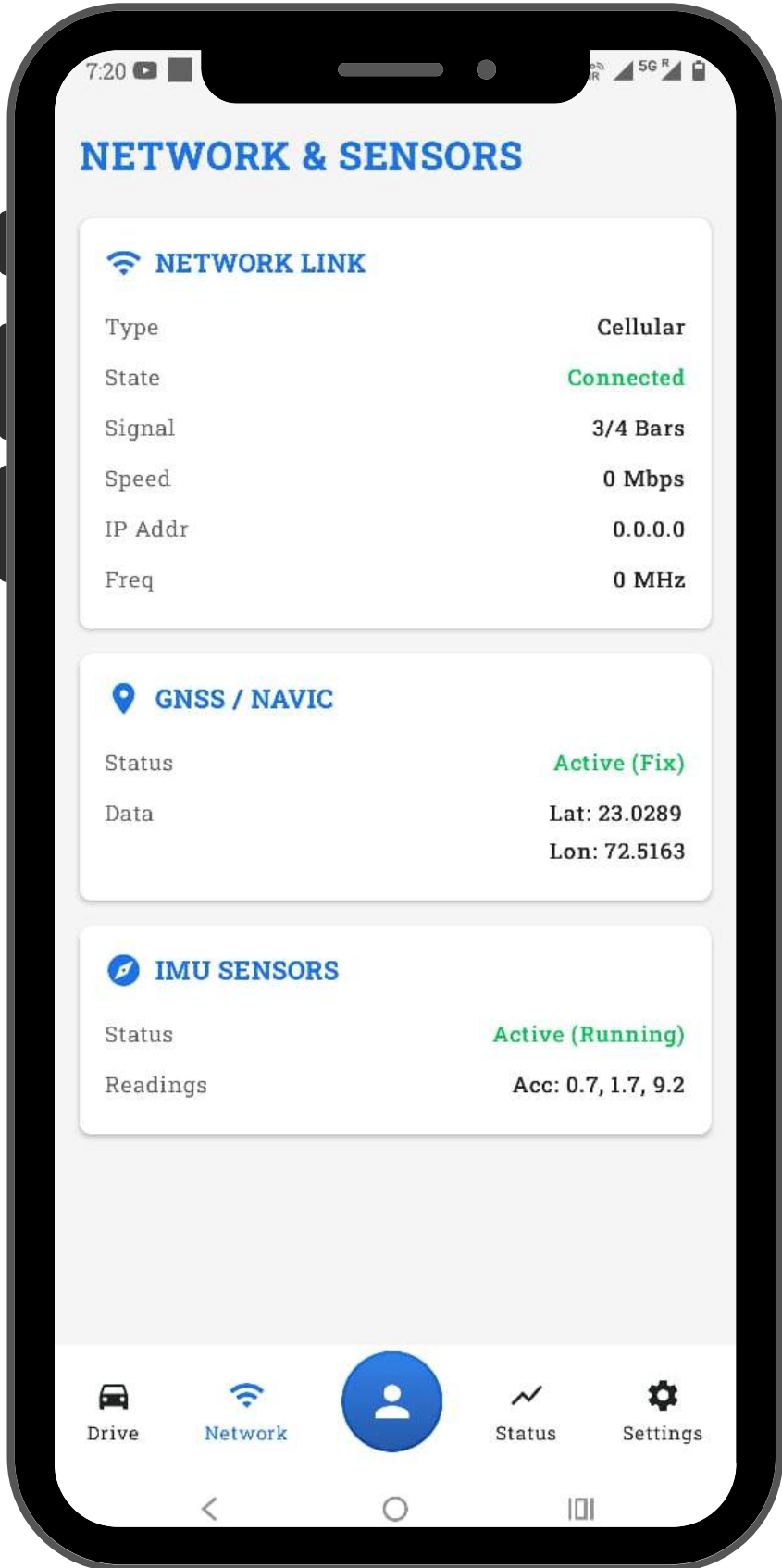
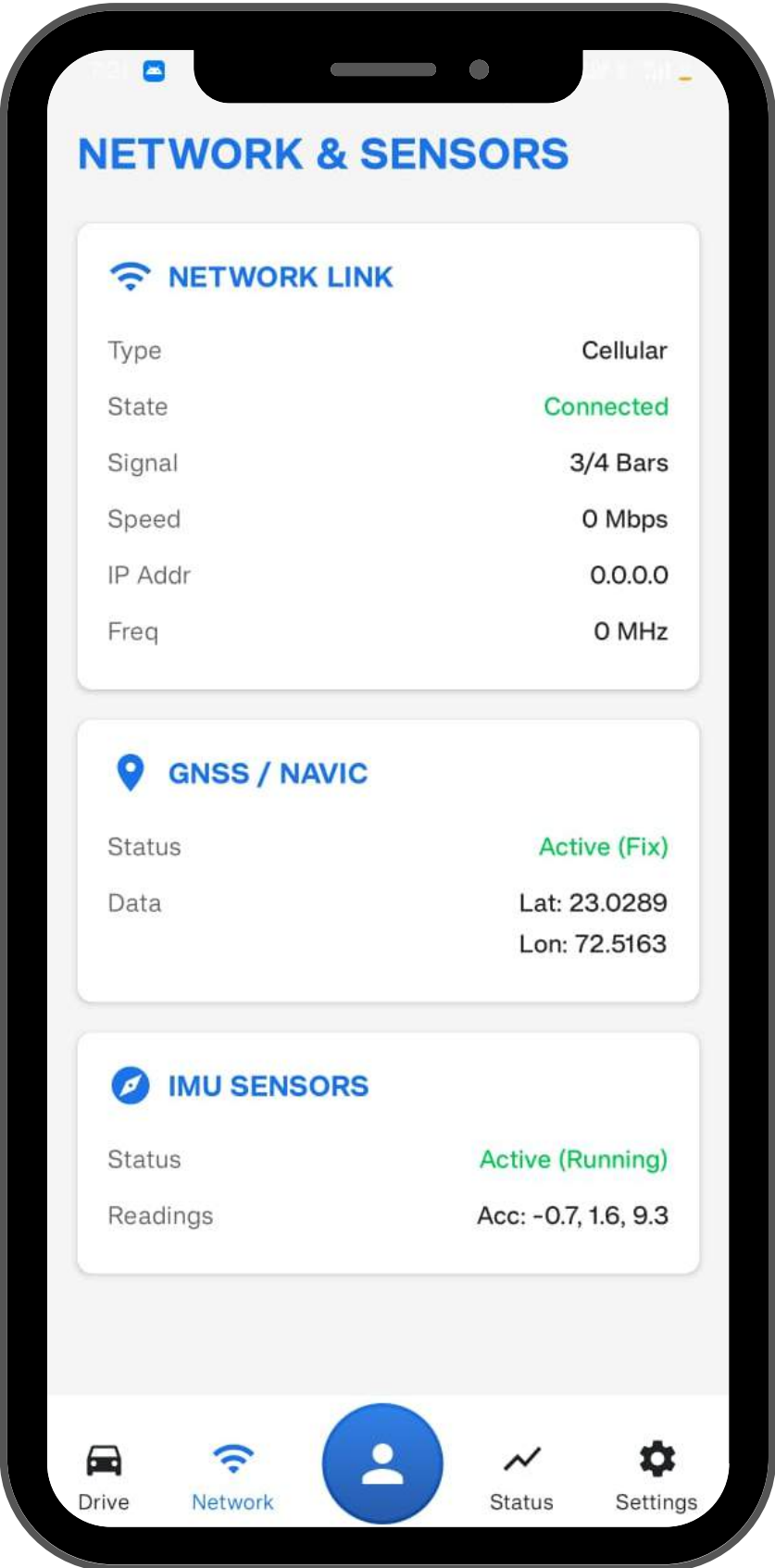
Component	Details	Per-User Cost (USD)
Open Street Map	Tile usage, trips, map visualization, caching	\$0.00
Protobuf	Lightweight packets, faster sync	\$0.02
Network Infra (P2P / Hosting)	Signaling + logging, always in sync.	\$0.38
Maintenance / Updates	Distribution, backend upkeep, bug fixes	\$0.14
Total cost :\$0.54		

AI + Smartphone Sensors + Real-Time Alerts = Safer Journeys

Smart Driving Safer Journey

LET'S MAKE DRIVING SAFER, TOGETHER
Our solution is scalable, secure, and ready for pilot deployment

*Thank
you*

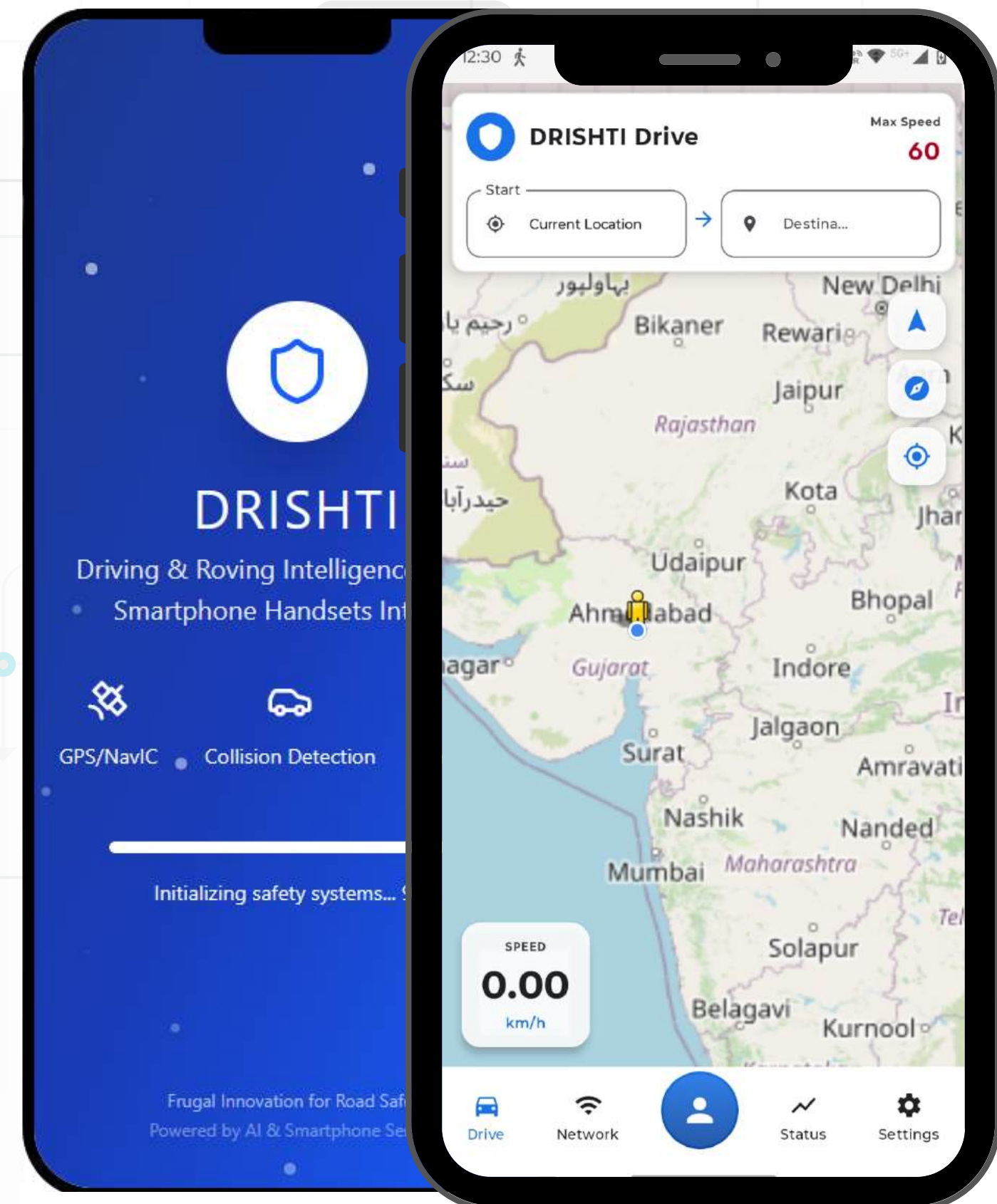


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FEATURES

EKF

EKF is good for smooth motion but reacts slowly to sudden changes, so we switch to IMU during fast movements.

Sensor Fusion (GPS + IMU)

GPS is slow but global, IMU is fast but drifts—combining them gives smoother and more reliable positioning.

GPS Spike Filtering

We collect real GPS data and remove sudden spikes using filtering and fusion because raw GPS isn't always trustworthy

Data Sharing

Devices share position and speed to build a cooperative safety bubble, improving relative accuracy even if one GPS signal is weak

Differential Positioning

Standard GPS isn't enough, so our long-term goal is to use nearby devices to push accuracy toward ~1 m without extra hardware.

Accuracy & Detection Probability

Accuracy is vital—alerts trigger only when both the risk and our confidence level are high.

Our App

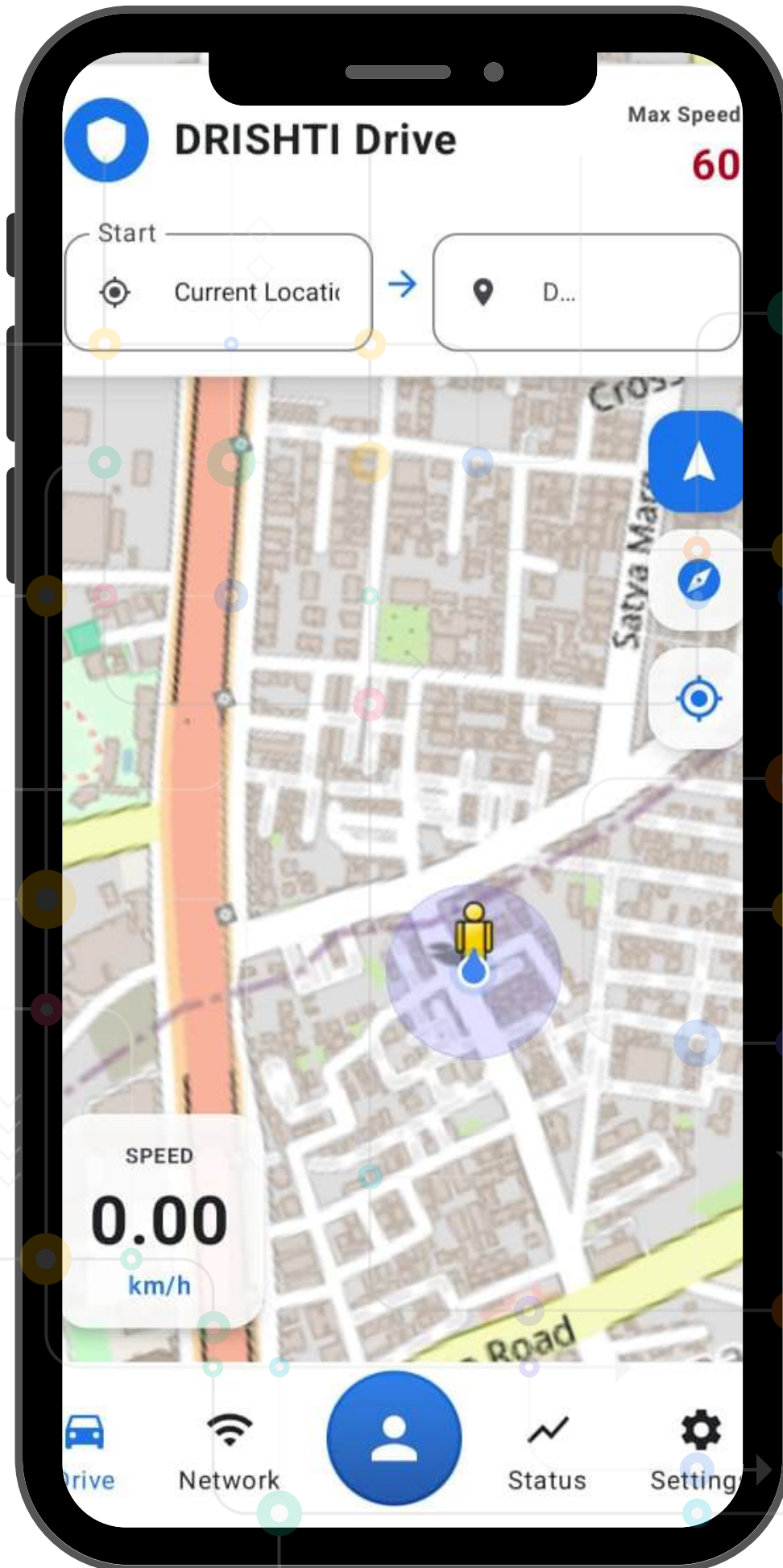


Connected using Wi-Fi on Mobile and Wi-Fi hotspot on Laptop

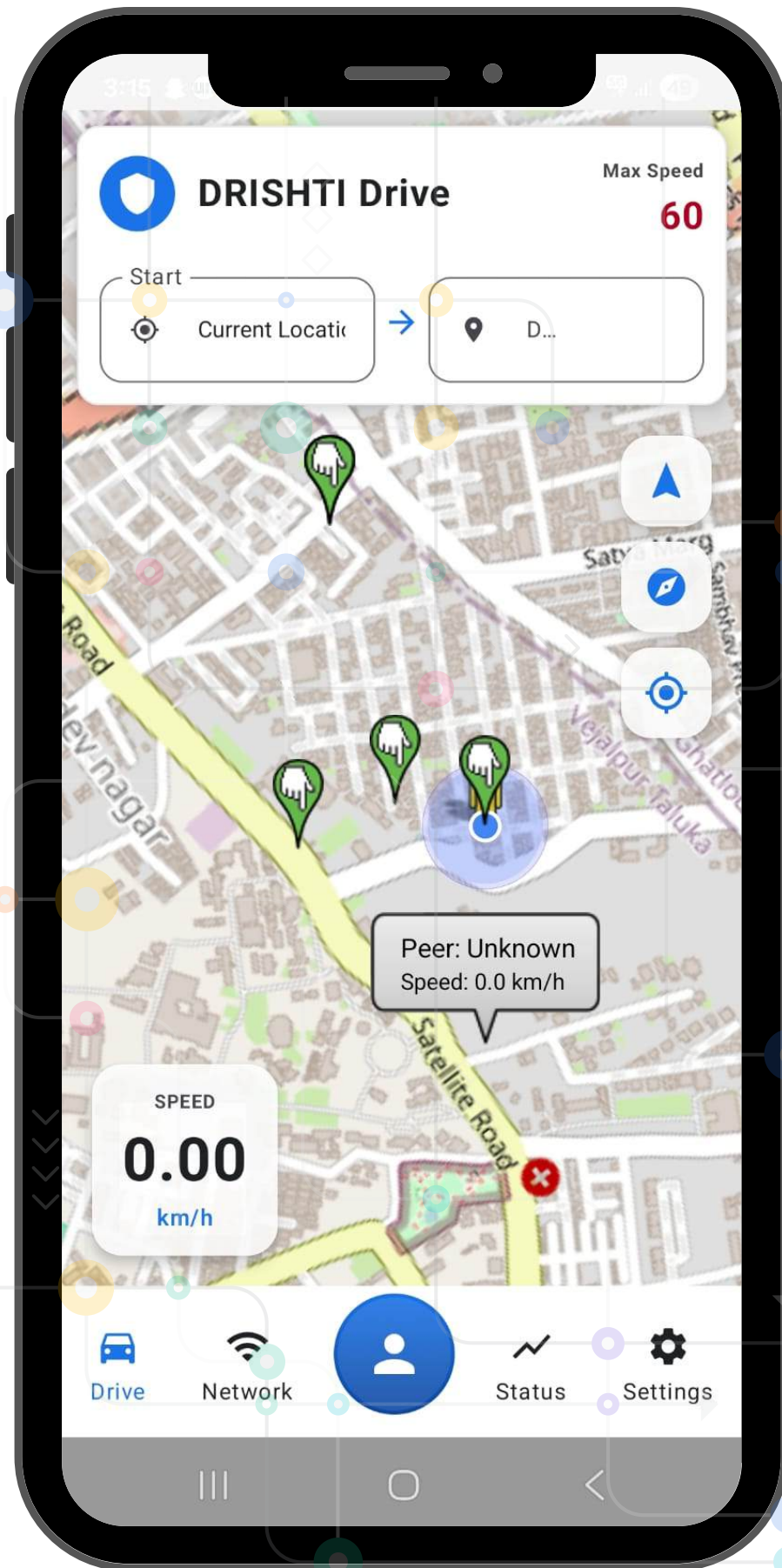
Static Condition

Position of user is seen

If any object comes closer to static object at a speed less than 5KM/H then no collision will be detected



Our App



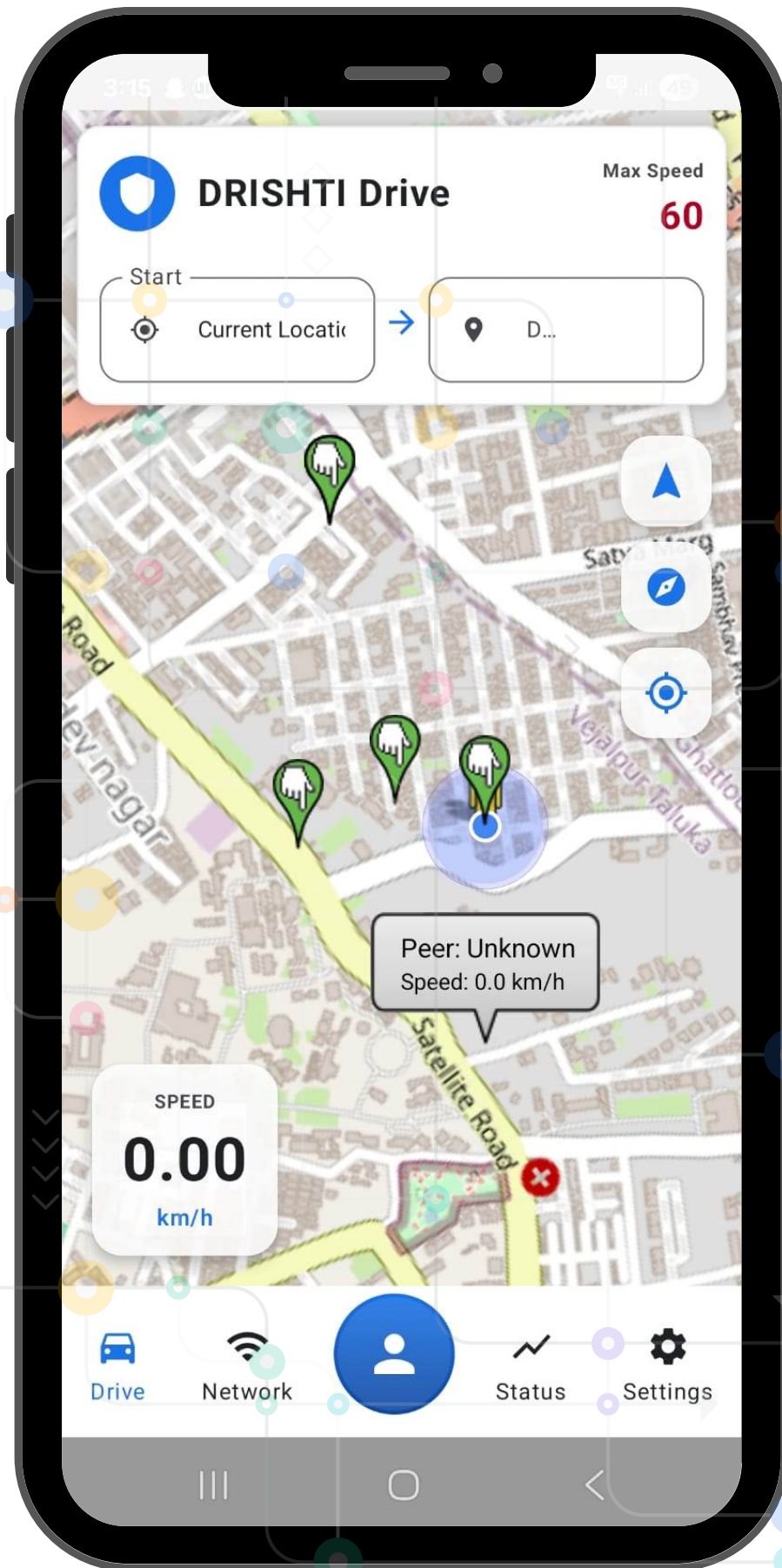
Dynamic Condition

If two or more objects come closer to each other at $V > 5$ KM/H then collision is detected

For Demo Purpose

Everyone in rang sees the alert

Our App



Lane Detection

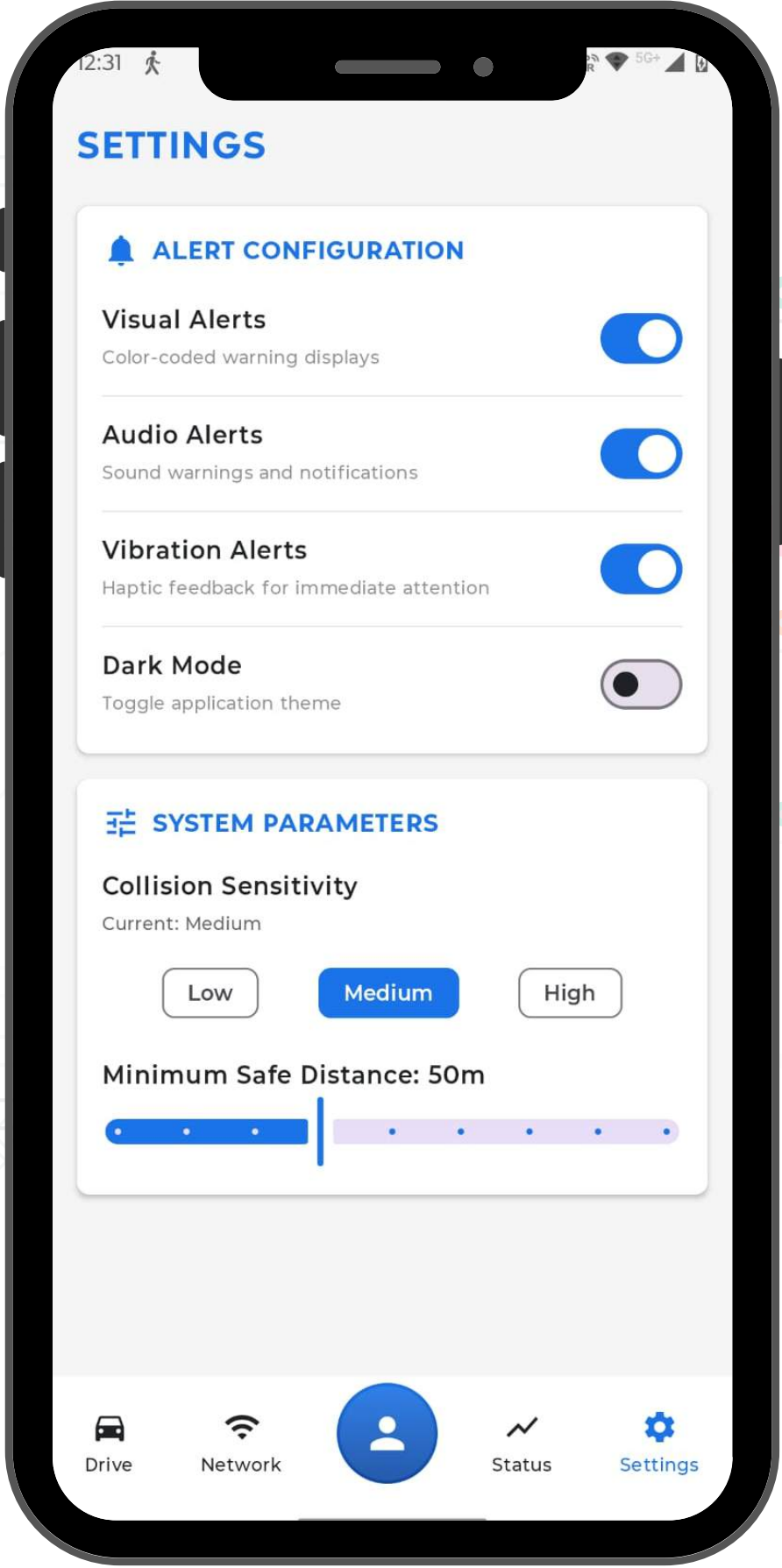
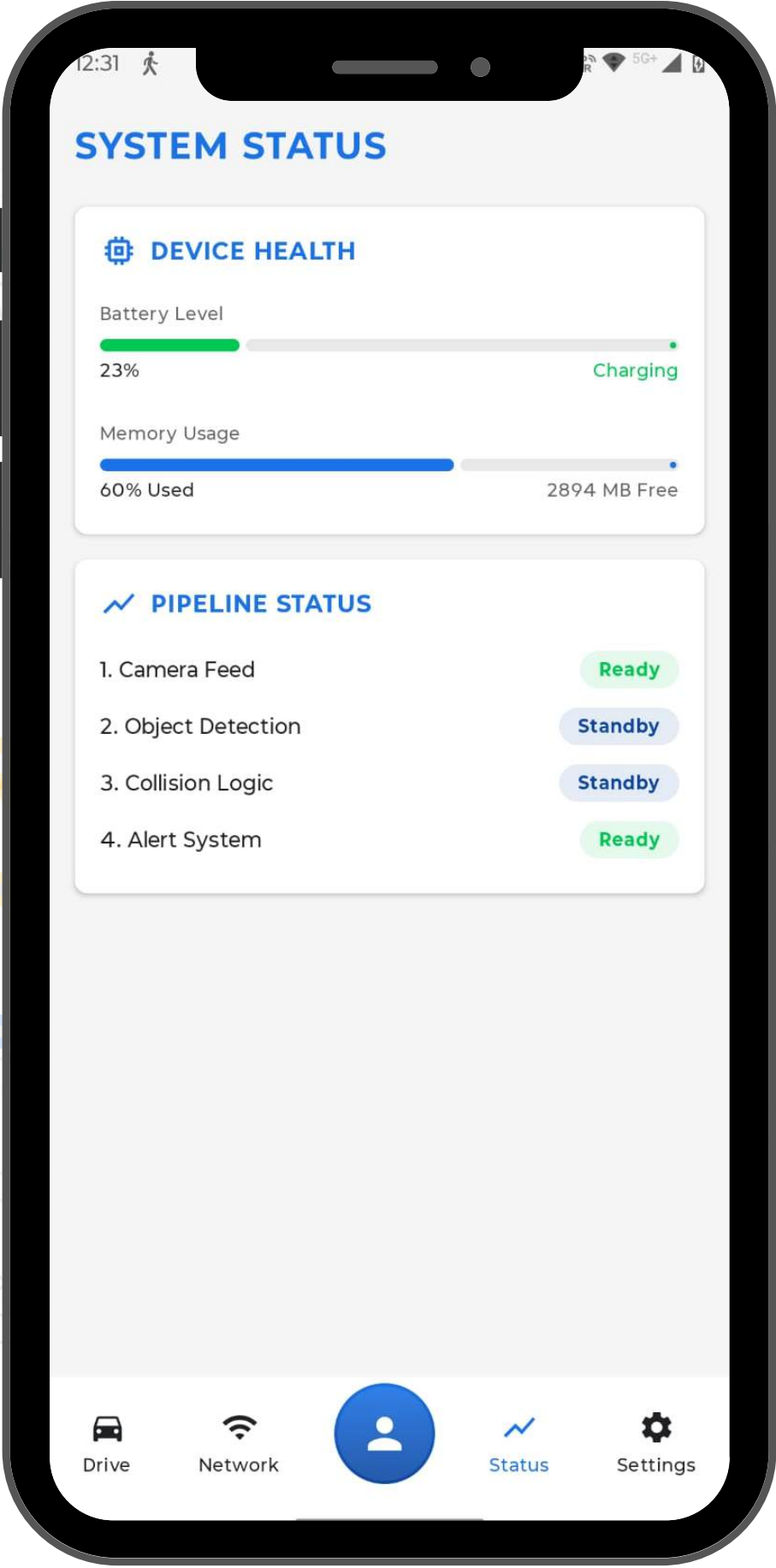
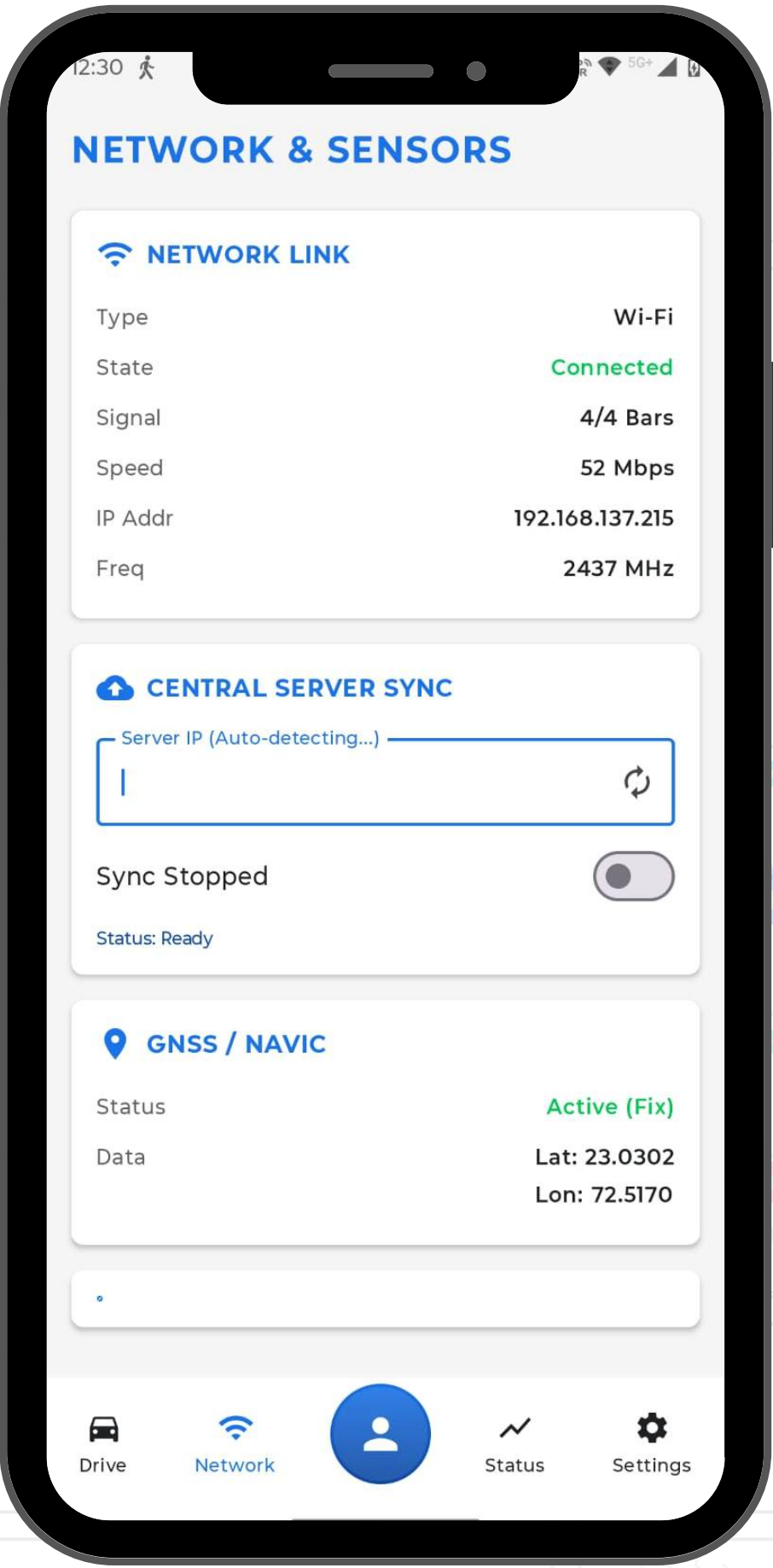
there is a safe distance
between objects

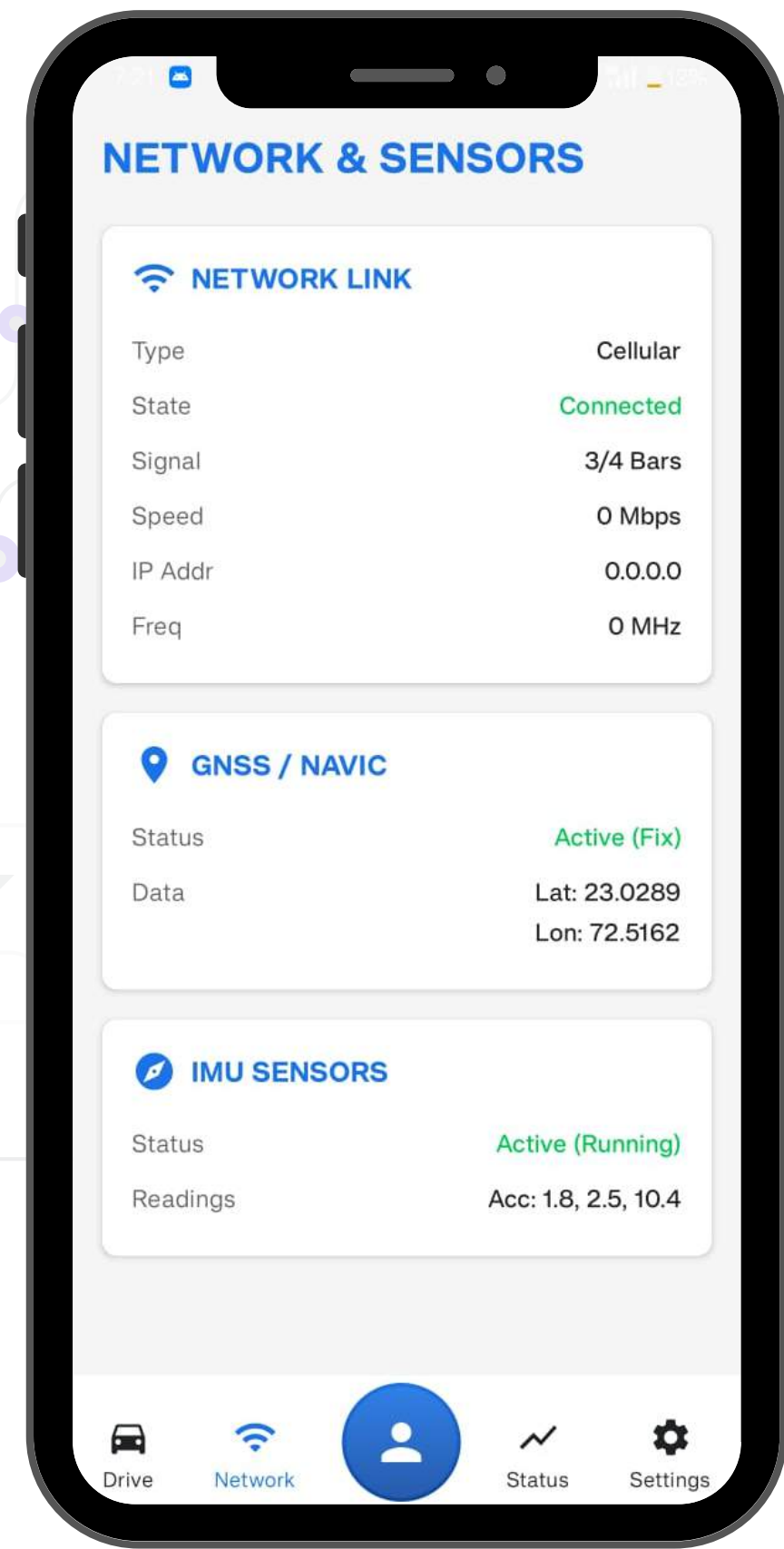
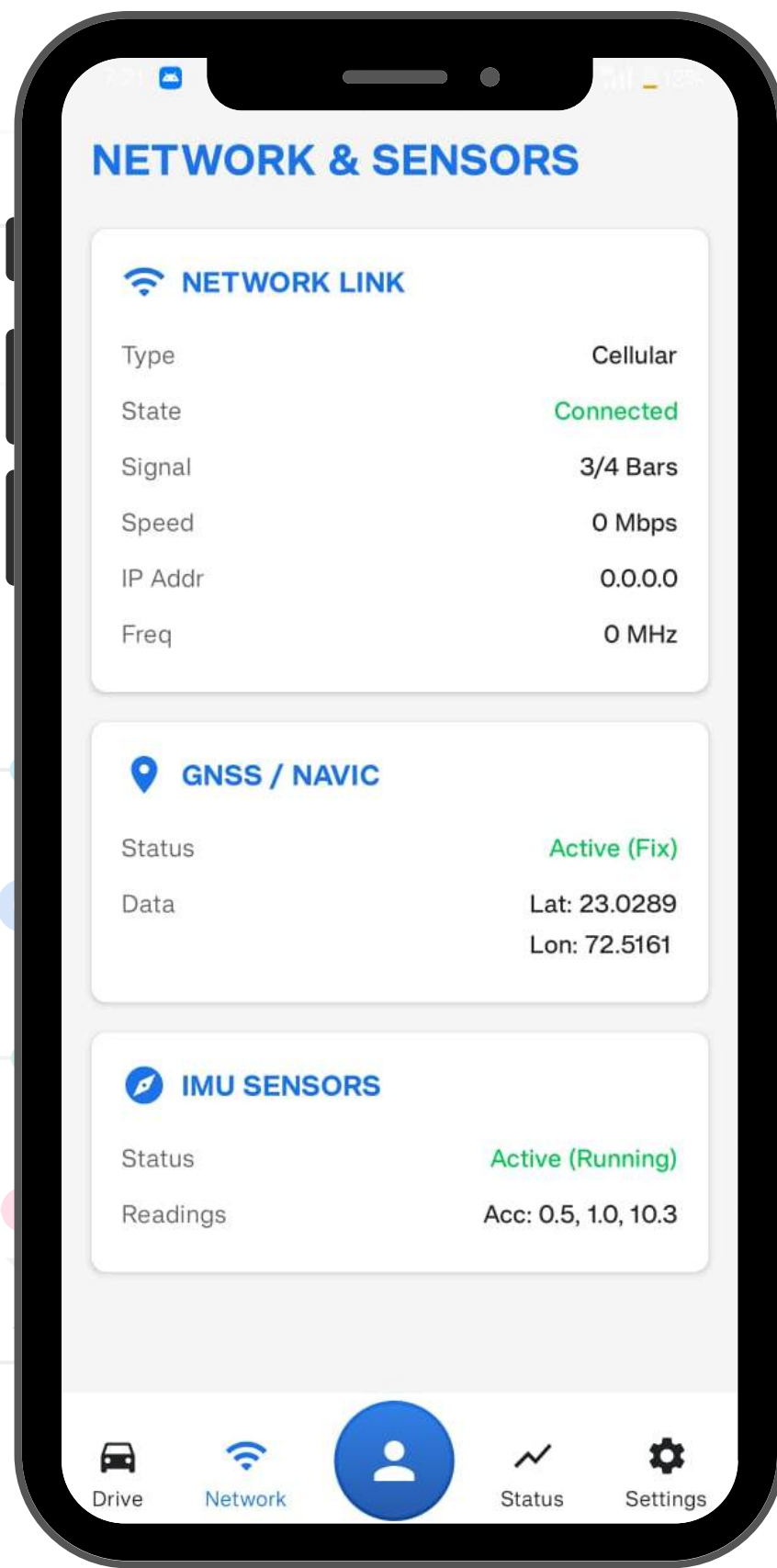
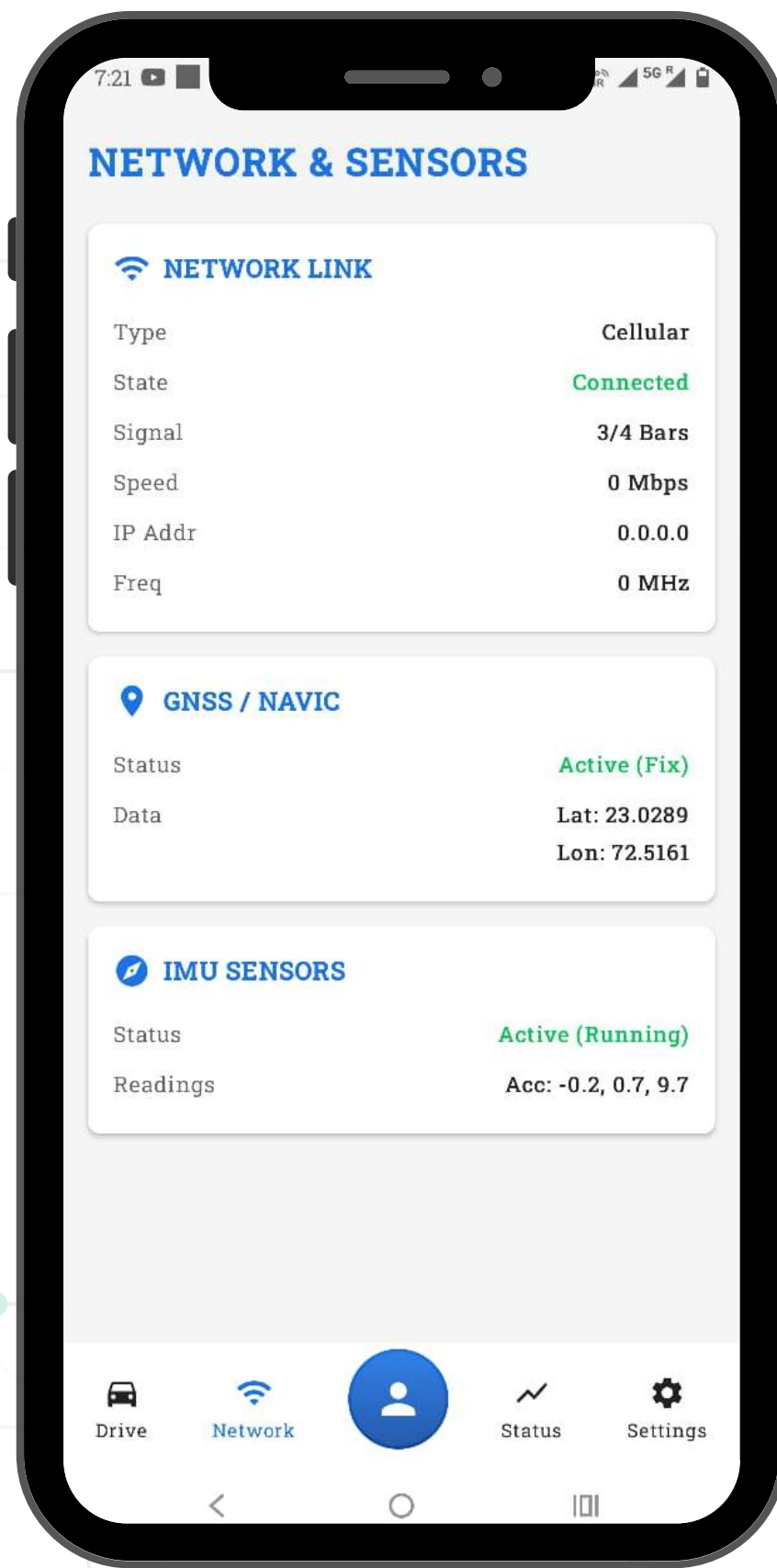
Our App



Collision Detection

This is the alert we see when collision happens





TEAM ROLES

Prashant Ranjan — Team Lead & Backend

- Leads project direction and backend architecture

Karan Sharma — Backend & App Integration

- Builds backend logic and connects it with the app

Ayush Pratap Singh — Research & Testing

- Research, real-world testing, and system validation

Ritik Prajapati — Server Load & Frontend Integration

Manages server load and connects backend results to UI

Sudeeksha Tripathi — Computer Vision (YOLOv8 + Depth)

Handles object detection, depth models, and AI vision

Prateek Pandey — UI Design & Maps

Designs UI and integrates map features

Our Team Members



Prashant Ranjan
24mc3035



Ritik Prajapati
24mc3040



AYUSH PRATAP SINGH
23CE3010



Sudeeksha Tripathi
24mc3050



Prateek Pandey
24mc3036



Karan Sharma
25CD3014